

PDF MAGICK ©

High Performance PDF Manipulation and Printing Software



*“Nothing is more powerful
than an idea whose time has come...”*

Quickstart and User Guide

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Installation

This section guides in the first time installation.

- Extract the contents of the ZIP file provided to a local directory.
- Create a shortcut for **PDFEDITOR.EXE** on your Desktop, Startmenu or Quicklaunch bar as desired. You can rename it to “PDF Magick”

Install Mandatory Components :

Microsoft .NET 4.0 Full Framework must be installed. If you attempt to start the application and are able to see the editor interface it is already installed and no action is needed. Otherwise see subfolder /Components/dotNetFx40_Full_x86_x64.exe or <http://www.microsoft.com/en-ca/download/details.aspx?id=17718>

Ghostscript 9.07 or higher is required for some operations.
<http://www.ghostscript.com/download/>

Adobe Acrobat is required for some operations.
The free Adobe Acrobat Reader is fine.
<http://get.adobe.com/reader/>

Install Optional Components :

Acroplot Repro. Only required if high performance PDF to Image conversion is desired. This component must be purchased and licenced seperately via <http://www.cadzation.com/downloads.htm>
Subfolder /Components/ acroplotreprosetup.exe has a 30 day trial version.

GraphicsMagick. Only required if “GM” Preview renderer will be used. There are 4 other preview rendering engines available. GM is recommended as the last option only if all others fail.
Subfolder /Components/ GraphicsMagick-1.3.13-Q8-windows-dll.exe

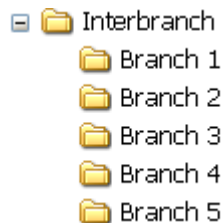
Adjust Configuration File (Optional) :

Open file **pdfeditor.exe.config** with Notepad or any XML editor and review section at the bottom. Modify the values as described below and use this updated configuration file when installing the software on multiple stations to suit your internal network.

```
<applicationSettings>
  <PDFEditor.My.MySettings>
    <setting name="AutoUpdatePath" serializeAs="String">
      <value>\\servername\sharename\PDF_Magick\</value>
    </setting>
    <setting name="Queue_IB_Root" serializeAs="String">
      <value>\\servername\sharename\Interbranch\</value>
    </setting>
    <setting name="Mailserver" serializeAs="String">
      <value>192.168.1.1</value>
    </setting>
    <setting name="LicencePath" serializeAs="String">
      <value>\\servername\sharename\PDF_Magick_Licence\</value>
    </setting>
  </PDFEditor.My.MySettings>
</applicationSettings>
```

AutoUpdatePath: Optional. Leave blank to disable updates. Otherwise the application will check for a current version from the given path. A network administrator can place a new version of the application into a central location and all stations within the network will update with no user intervention.

Queue_IB_Root: Optional. Point to a shared folder that all branches, departments have access to. It is used in the “Send Job” Screen as a path with subfolders. You can create any number of subfolders within /Interbranch/ to suit your needs.



Mailserver and AdminEmail: Optional. Internal Email server to use to alert administrator that a user has requested a new Licence.

LicencePath: Optional. When a user “Requests” an activation licence, the licence file (.LIC) will be placed into this path. A network administrator can retrieve the .lic from this path, activate and replace back into this folder. The program will attempt to “Authorize” the licence from this path if “Authorize” is clicked within the Licence Manager.

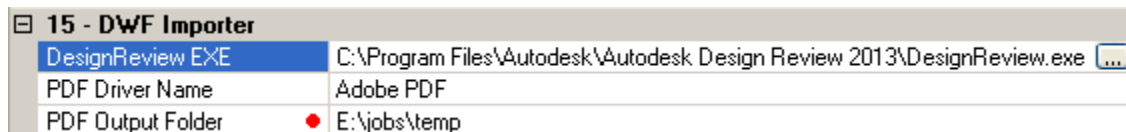
The active licence file is always in the application directory, regardless of the network path.

Install DWF Converter (Optional) :

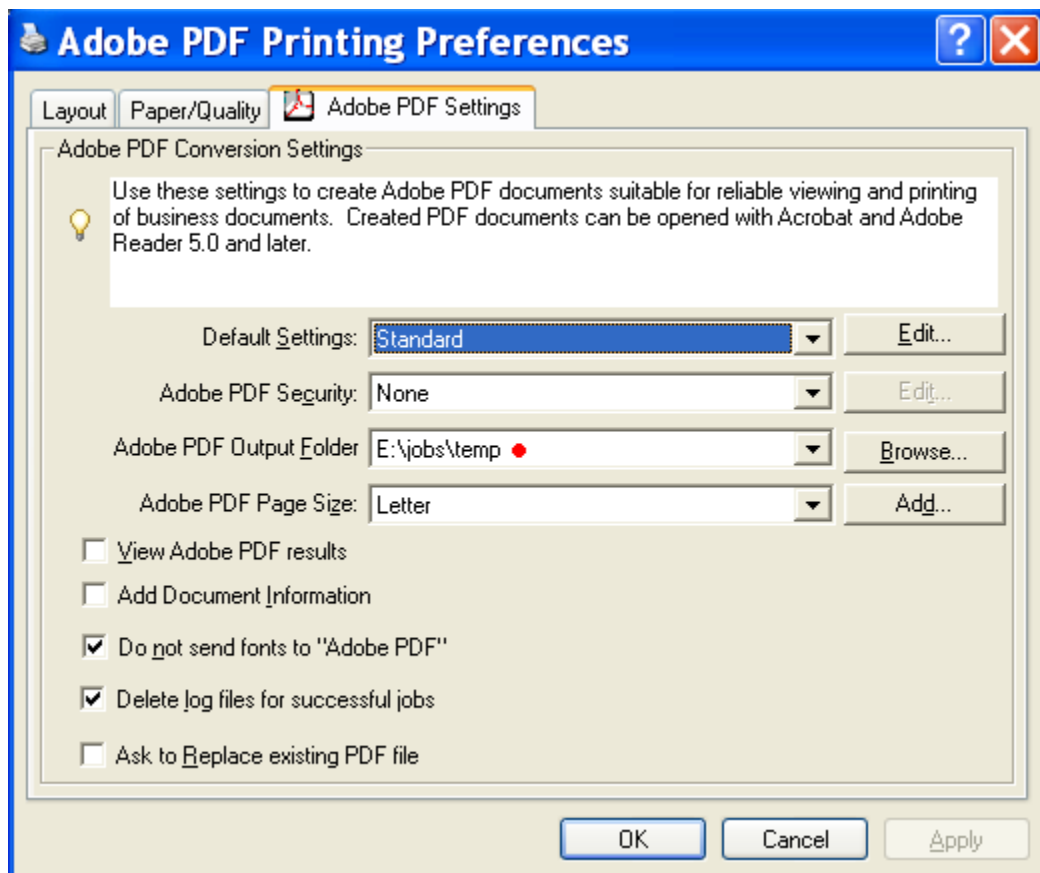
If you want to work with DWF & DWFX files you can install and configure a converter that will do vector based DWF to PDF Conversion. This converter preserves the original quality of a vector DWF and is immune to quality loss when enlargements or reductions are made. There are 2 requirements for this Converter :

- Adobe Acrobat PDF Driver (The windows system driver that comes with Acrobat Professional)
- Autodesk Designreview 2013 (Free software available at <http://usa.autodesk.com/design-review/download/>)

Once the above components are ready you need to make some configuration adjustments in PDF Magick.

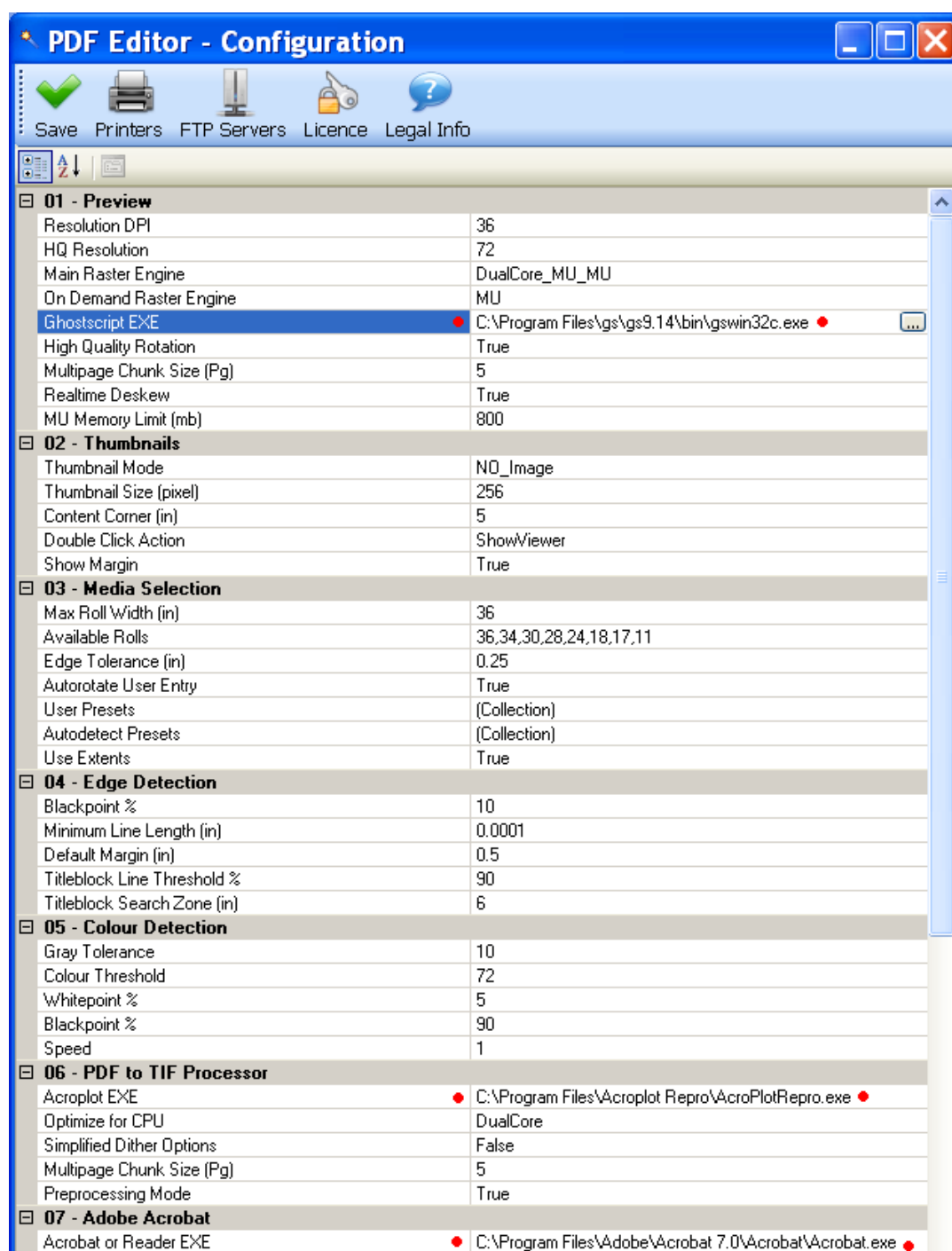


Finally the “Properties” of the “Adobe PDF” Driver must be adjusted.



First Startup :

At first startup the application will attempt to detect and configure installed components. In case you have multiple versions of those applications and want to use a specific version or if the automatic detection fails you can adjust the launch path for the items marked in red inside the Configuration settings.



Expectations & Capabilities

This section explains what PDF Magick is designed for and does very well.

PDF Magick is not a replacement for Adobe Acrobat or Enfocus Pitstop.

PDF Magick has features that will outperform the above mentioned or similar software by far, but it is not a replacement of all features of those applications.

What is PDF Magick designed for:

- Preview & manipulate size, sequence, alignment, orientation of PDF files, in batches or individually.
- Prepare for Printing or Electronic manipulation and export only.
- Print on Largeformat & Smallformat devices, Black&White or Colour – regardless of manufacturer.
- Extremely productive! It will outperform any other available software on the core tasks.

If you are processing files and used to working with Reprodesk, PLP, Metaprint, KIP Request or similar software you will find that PDF Magick processing speed will outperform that software in BW or Colour by a margin of 30:1 (3000% faster) or more when working in native PDF Mode and by 2:1 (200% faster) or more when working in Image mode.

“Simplicity is the ultimate form of sophistication...”
Leonardo da Vinci

Main Features:

- Preview PDF files with high performance WYSIWYG viewer – full image, zoom, or thumbnail/edge.
- Adjust Sequence, Orientation, Zoom, Alignment and Papersize – in batches or individual ly with real time preview.
- Edge & Titleblock detection for precise auto-alignment.
- Grouping of pages for sections, disciplines within a job. Manual or Automated.
- Multiple and unlimited “versions” (Fullsize, Halfsize, 11x17, A0, A3,...) using same files and within same job.
- No propriatery file formats. All files remain as PDF in open directories.
- Flexible configuration for automatic media selection based on common media roll width, sheet sizes, metric or imperial units.
- No processing, imaging or database “server” requirements. Standalone & self contained application.

Advanced Features:

- Colour & Redline content detection.
- Overlay & Watermark style stamping, with variable content.
- Annotation & Font embedding.
- Support for automation of doublesided document setup, blank page insertion.
- Vector Linethickness override.
- Brightness and Colour Saturation controls for vector Grayscale & Colour content.
- Crop pages.
- Imposition capability with unlimited N-Up, advanced layout and piece repeat options, cropmarks, bleeds, dielines.
- Automatic Tiled (side by side) and/or stacked, 2 up or 4 up printing.
- Deskew using any rotation degrees with on-screen detection.

- Enforce document management rules for 3rd party planroom indexing.
- OCR Based Renaming with template matching to adjust for image shifting.
- Horizontal and Vertical Rulers with configurable Guidelines in viewer.
- Drag Drop files across multiple directories, with automatic grouping and slipsheeting.
- Job Summary for quicker quoting or billing, separated by Small-, Largeformat, BW, Colour.
- Task based workflow for small locations. (Add Files > Setup > Print)
- Queue based workflow for large operations (Add Files > Setup > Send To Queue > Autorefresh Queue > Print from Queue). A queue is a simple collection of windows system subfolders. No propriatery queue formats.
- Interbranch Job Transfer capability via network share, ZIP files or FTP.

Output Options:

- Export to vector PDF. Single or Multipage, export to device “Hotfolder”.
- Convert to raster TIF or JPG, with custom dither options. BW, Grayscale or Colour mode.
- Print to Largeformat or Smallformat devices, BW or Colour – all from the same interface & same job setup. No restrictions on manufacturer. No restrictions on number of printers.
- Print history available if job was Queue based.

Print Device Requirements:

PDF Magick is designed completely device agnostic. The job editor and processing engines have no dependency or knowledge of any manufacturer or device. The print submission and device configuration options have multiple **submission protocols** that can be configured depending on the device and its capabilities. It is possible to configure the same device via multiple options and different protocols.

In order for a print device to be compatible with PDF Magick **any one** of the following conditions must be met.

- The device is accessible by an IP address and supports LPR or FTP protocol.
- The device is accessible as a windows shared network printer.
- The device has a standard windows system driver, installed on the local PC system.

Benefits of Device Independent Design:

The device requirements are non-restrictive and allow almost any printer to be configured and used. There are great benefits and flexibility by this design for the users of the software.

- There is no need to purchase additional “print enablers” for specific devices that some vendors will try to sell.
- There is no need to worry about current or future devices being incompatible with PDF Magick. New devices will simply work as long as they satisfy the basic connectivity requirements explained in chapter 3 and follow the industry standard Adobe PDF Spec ISO32000.
- The software can be used to setup print ready or viewing ready & organized jobs and export the result to PDF files to give to other individuals or companies that do not have PDF Magick. They can view or print on their own devices. There are no propriatery file formats.

In order to use the program with its maximum performance when printing in PDF mode the device must have an onboard, native PDF/Postscript interpreter. On Largeformat devices this is usually a configuration or licencing setting enabled with a security key by the vendor.

Even if the device does not have native PDF/postscript capability, printing is still possible with 2 other methods; “Print as Image” or Printing using the local system driver for that device.

PDF Magick has been tested and works with Largeformat and Smallformat devices from HP, Canon/Oce, KIP, Xerox and any other manufacturer that meets the above basic requirements.



Configuration of Devices

This section explains how to add new devices, what the different submission protocol options are and when to use them.



Adding a new device:

Click “**Add Printer**”. A new line will appear in the list.

Highlight it and look into the “**Settings**” area of that device.

Start with the “**IP Address**”, ensure a network ping response is received if the device is turned on.

Select “**Device Model**”. This will prepopulate some default settings for this model. This setting is used for Device roll detection. The software will work with the devices own interface even if your exact model is not in the list. “Generic Smallformat” and “Generic Largeformat” options can be used at any time.

Give the device a name, “**Printer Name**”. This text will be visible to operators.

Select suitable “**Submission Protocol**” if device model was choosen “generic”. The submission protocol is the most important setting. Please see next section about detailed explanation.

Select “**Page Submit Method**” – Singlepage is recommended for Largeformat devices; so that printing can start right away after the first page is processed on big jobs. Multipage is recommended for Smallformat devices or when the submission proctol relies on Acrobat and the windows system driver.

Review and adjust remainder of options as needed. Section 03 – Media / Rolls will be used only when a known largeformat device model is configured. It is ignored for smallformat sheet based devices.

Printer Configuration

Save

Close

Add Printer

Remove Printer

Print Test Page

Move Up

Printers

Printer Name	IP	Model
☒ KIP - Mono	192.168.11.40	KIP7800
☒ CADD HP LaserJet	192.168.10.33	HP_LaserJet
☐ Xerox 470 Docutech	192.168.10.53	XEROX_DC_470
☒ Canon C3380	192.168.10.226	Generic_SmallFormat
☒ Xerox Phaser 7760	192.168.11.41	XEROX_Phaser_7760
☒ DC252 Colour	192.168.11.44	Generic_SmallFormat
☒ Xerox 4127	192.168.11.196	Generic_SmallFormat
☒ Oce #1	192.168.10.169	OCE_TDS860_Fw1013
☒ Oce #2	192.168.10.160	OCE_TDS860_Fw1013
☒ Oce #3	192.168.10.117	OCE_TDS860_Fw1013
☒ Oce #1 LPR	192.168.11.209	OCE_TDS860_Fw1013

Settings

01 - Network

IP Address	192.168.10.229
Submission Protocol	LPR
LPR Queue Name	LPQUEUE
Network Share Name	
Driver Name	

02 - Printer

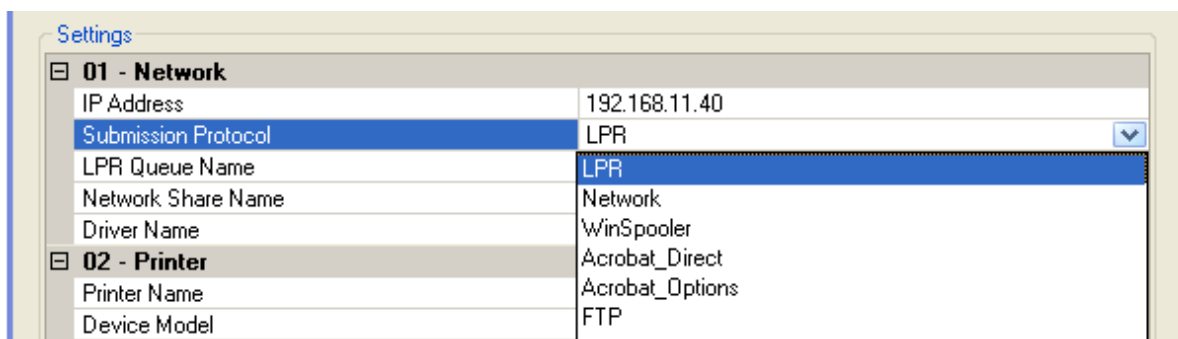
Printer Name	Colorwave 1
Device Model	OCE_CW600
Page Submit Method	MultiPage
Reverse Collation	False
Active	True
Refresh Interval	0
Print while Processing	False
Ocs TDS Logging	False

03 - Media

Number of Rolls	4
Read Media Status	True
Warn if exact width is unavailable.	False

IP Address

Submission Protocols:



The screenshot shows a 'Settings' window with two main sections: '01 - Network' and '02 - Printer'. In the '01 - Network' section, the 'Submission Protocol' is set to 'LPR'. In the '02 - Printer' section, the 'Printer Name' is set to 'Acrobat_Direct' and the 'Device Model' is set to 'FTP'.

Settings	
01 - Network	
IP Address	192.168.11.40
Submission Protocol	LPR
LPR Queue Name	LPR
Network Share Name	Network
Driver Name	WinSpooler
02 - Printer	
Printer Name	Acrobat_Direct
Device Model	FTP

LPR :

This is the preferred and most efficient protocol for largeformat devices that have native PDF capability. The device must accept LPR protocol, please check its documentation how to enable it on the controller of the device.

If your PC station is Windows Vista, 7 or 8 please ensure the LPR subsystem is enabled on your PC. It is disabled on those windows versions by default. This can be done through "Control Panel/Programs and Features/Turn Windows features on or off". In the dialog expand "Print and Document Services." Check "LPR Port Monitor."

LPR Queue name setting is required on some devices. The default of RAW or LPQUEUE will work fine in most cases. For KIP class devices using IPS1, IPS2, IPS3 and configuring those queues with special print or colour settings is possible.

Specific print and device defaults, like media selection and roll switch policy need to be configured on the controller or the interface of the device.

NETWORK :

This is for networked printers that have native PDF capability. **Network Share Name** is critical and must be the name of the shared printer. The files will be submitted to [\\IP\Network Share Name](http://IP/Network Share Name)

It is recommended for largeformat devices if LPR is unavailable.

Specific print and device defaults, like media selection and roll switch policy need to be configured on the controller or the interface of the device.

FTP :

Not recommended unless LPR and Network are unavailable. The device must support standard FTP protocol on port 21 with anonymous login. The device must have native PDF capability. **Network Share Name** is critical and must be the subfolder of the receiving FTP server. Files will be submitted to <ftp://IP/Network Share Name>

Specific print and device defaults, like media selection and roll switch policy need to be configured on the controller or the interface of the device.

WINSPoolER :

Not recommended unless other options are unavailable. The files will be submitted to the standard windows spooling system. A local driver must be installed and **Driver Name** defined. The device defaults will be used as print settings and no per job configuration of settings is possible unless the device defaults are changed on the device itself. The device must have native PDF capability.

ACROBAT_DIRECT :

This is the recommended protocol for basic smallformat devices that require no advanced print setting configuration and no finishing or media selection options per job. A basic office desktop smallformat printer is an example. The device defaults will be used for print settings. There won't be any dialog boxes when printing. Page submission method is recommended to be **Multipage**.

A local driver must be installed and **Driver Name** defined. It does not matter if the device is PDF capable or not.

ACROBAT_OPTIONS :

This is the recommended protocol for advanced smallformat or largeformat devices where the user must have control of every setting of the device for media selection, duplex printing, colour settings, finishing options etc., or the device is not natively capable of PDF/Postscript format files.

PDF Magick will create a print ready file and pass it to the Acrobat Print driver interface. Number of sets in PDF Magick will be passed into the driver interface. Page submission method is recommended to be **Multipage**.

A local driver must be installed and **Driver Name** defined. It does not matter if the device is PDF capable or not. This option is technically the same as if printing a PDF with Adobe Acrobat.

Singlepage or Multipage Submission:

There is no general answer which one is better over the other, that will cover every scenario and device. It is better to understand the pro's and con's of each and choose the method most suitable. Some controllers & onboard rips work better with one or the other.



In **Singlepage mode** and when "Print while processing" is enabled, the software submits 1 page at a time to the controller as soon as it is ready. Printing starts almost instantly even for jobs with 1,000 pages. LPR, FTP, Network and Winspooler protocols are recommended in this mode. You will see **1 entry per Page** in the queue of the print device.



In **Multipage mode** the software creates a package for all pages first and submits in sets. The device needs to process that package first. On some devices starting to print may take much longer, but once it is processed printing may continue much faster then processing each page one by one. The Device queue will show **1 entry per Set** if Protocol is LPR, FTP, Network, Winspooler or Acrobat_Direct and **1 entry per Submission** if Protocol is Acrobat_Options.

Singlepage or Multipage Recommendations:



Prefer Singlepage if the device is Largeformat Black & White and there is no risk multiple users from different stations can submit at the same time to the same device.



TIP: Review your workflow procedures and job scheduling if you have conflicting jobs on the same machine at the same time from different users or consider implementing a “**Queue Based**” workflow instead of a “**Task Based**” workflow which is supported in PDF Magick as well. See chapter 7 schematics about workflow options.



Prefer Multipage if the device is Largeformat Colour and you want to see a single entry in the device queue, for easier tracking or cancellation.



Prefer Multipage if the device is Smallformat, BW or Colour. Combine this with Acrobat_Options Protocol to control all of the device capabilities via its driver interface.

Quick Start

This section explains the basic sequence of steps to initiate a new job, make some modifications and export the results to a file or printer. Please refer to the function and command reference or the training video's for further detailed information about all commands and options.



Adding Files:

Click on “New Job” to clear any previous items in the editor before starting a new job.

“Drag and Drop” from Windows Explorer is the easiest way to add new files. New files will always be added to the bottom of the job if done via Drag and Drop. You can add files or directories. If there are subdirectories their contents will be added as well. If the subdirectories have the same filename they will be renamed in place with a ~1 ~2 ~3 ... suffix. If you want to add files at a specific spot inside the existing job use the “Add Files” Toolbutton. New files will be added after your current selection position.

Immediately after dropping new files you will see new entries in the list in a light gray colour. The preview generation engine will engage and prepare viewable images of each file. Progress is shown by items changing to black colour. You do not need to wait for the preview generation to finish. You can make modifications on any column at any time. There are 4 different preview generation engines. You can adjust their settings in section 01 of the Config.

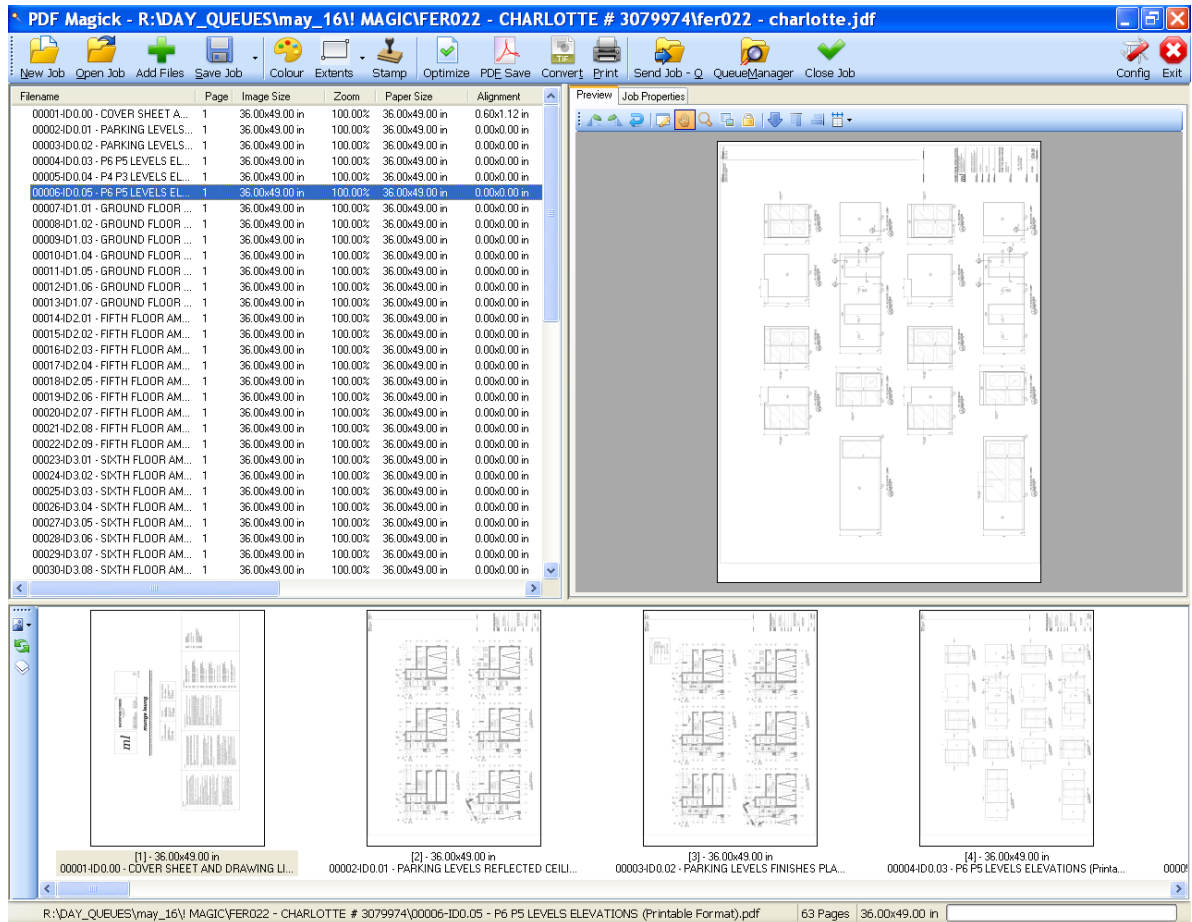
Making Modifications:

Right click the “Zoom”, “Papersize”, “Alignment” and “Rotation” columns and make adjustments as desired. Clicking on the Column Heading will apply selection to all files. Right click on one or more selected files will apply settings to the selection only. You can use “Copy All” or “Copy Down” with partial selections to quickly apply properties to multiple files.

Preview Fit:



Using the Alignment Mode Icon (Hand) you can view the full image. Mouse drag & drop will position the image on paper. The preview will turn off if more then 1 image is selected.



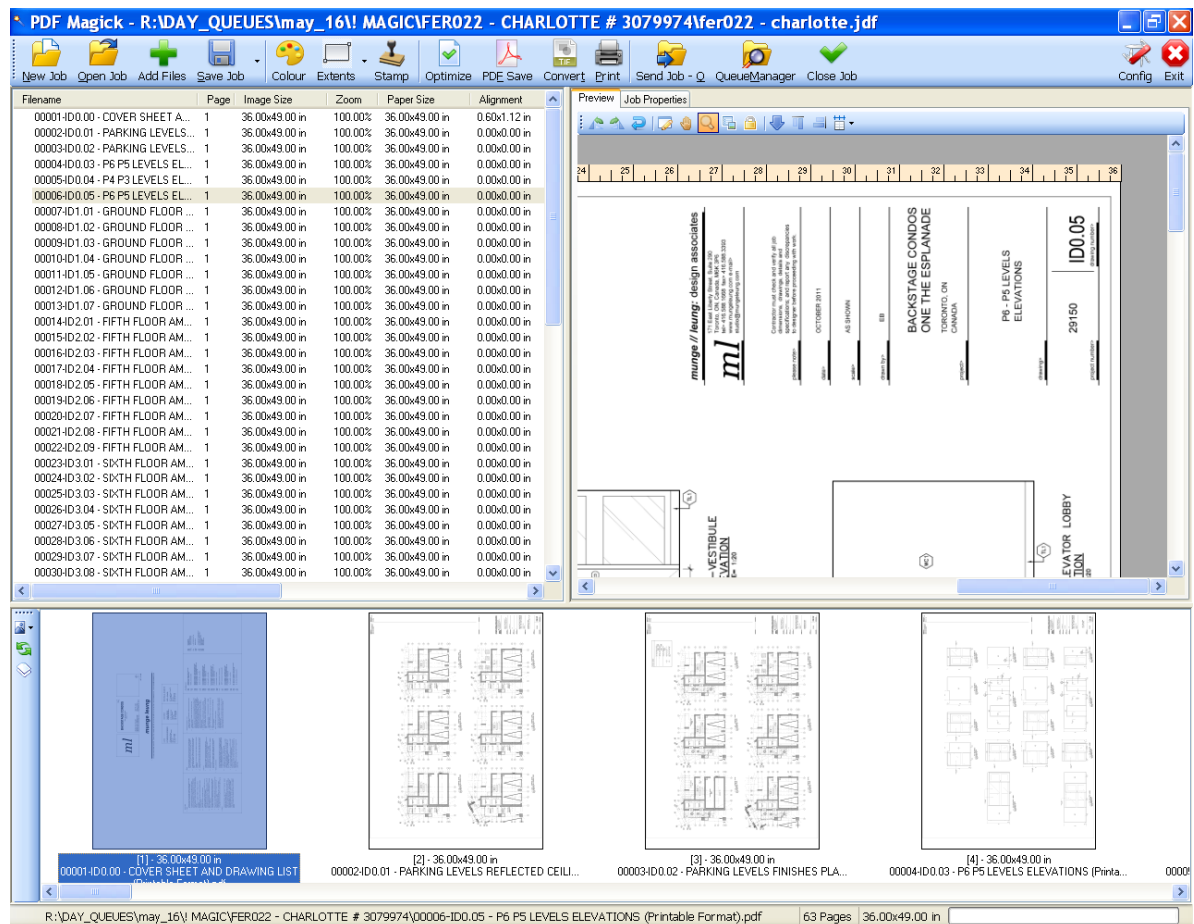
The bottom status bar of the editor shows currently selected filename with its full path, the total number of pages in the job and the current page size if 1 page is selected.

The bottom status bar will show the number of selected pages if it is more then 1 page allowing “counting” of chunks quickly. Experiment with automatic and manual “Grouping” to identify logical groups of files. Refer to the training video for a demo.

Some of the common commands like rotating and aligning are available at the top of the viewer as well for users that like mouse control. For convenience the user can also use arrow keys (up & down) to move between pages and (left & right) to rotate.

Preview Zoom:

With the Zoom Icon you can get a closer view to read drawing name and title. The resolution depends on your Config settings. Mouse drag & drop will position the image on paper. Mouse wheel will zoom further in and out of the existing preview but will not change the resolution.

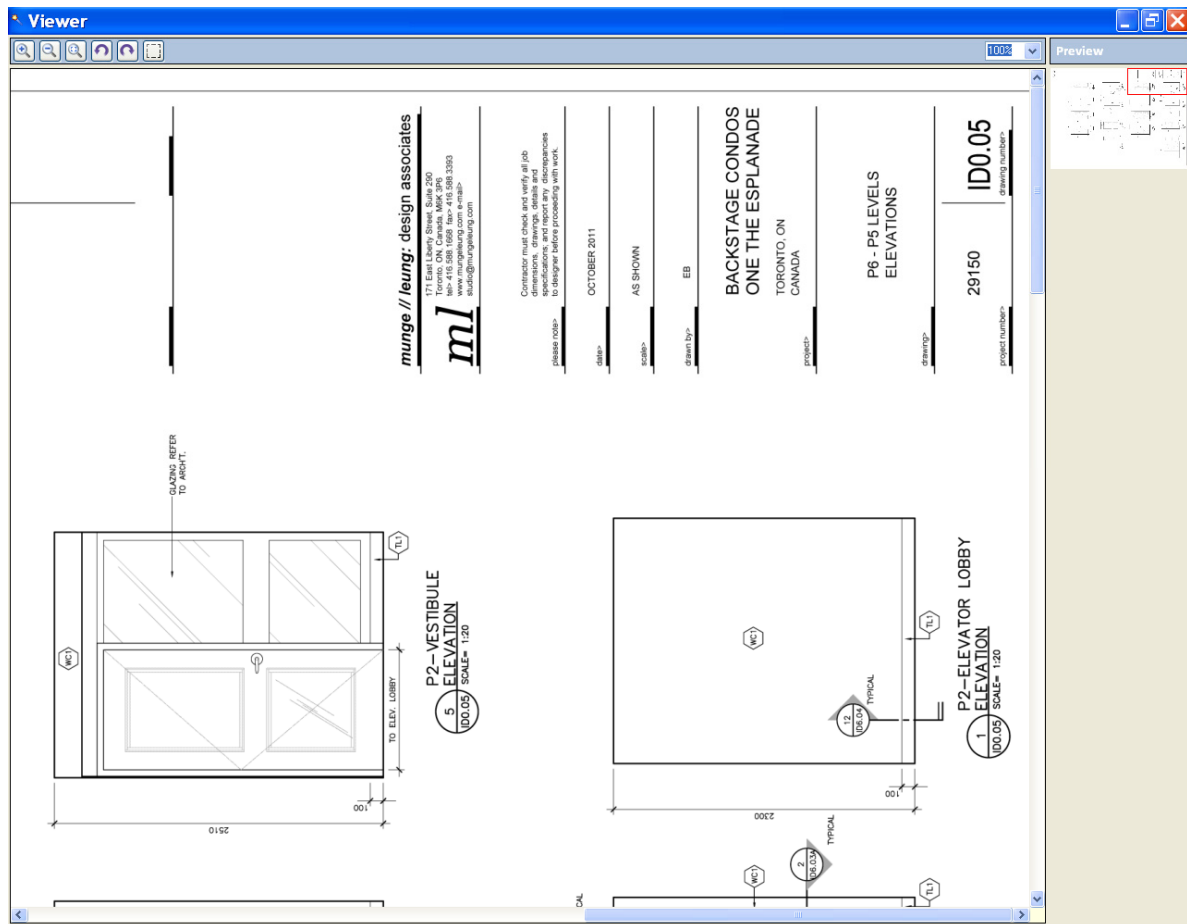


If your intention is too zoom in really close to read a small detail it is better to open the drawing in the native editor. The build in editor is designed for high speed preview and not to render at very high resolution.

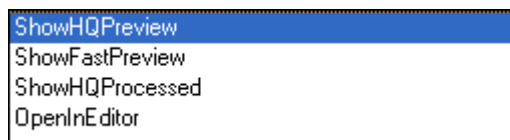
There are multiple ways the native file can be opened; via right click in the list view and select the appropriate command; the command icon on top of the viewer ; double click in the list and configure action accordingly; double click on thumbnail and configure action accordingly.

Preview Fullscreen:

Double clicking a file will open the full screen viewer, or the associated editor.



Depending on your selection in Config section 10 you can configure different double click actions.

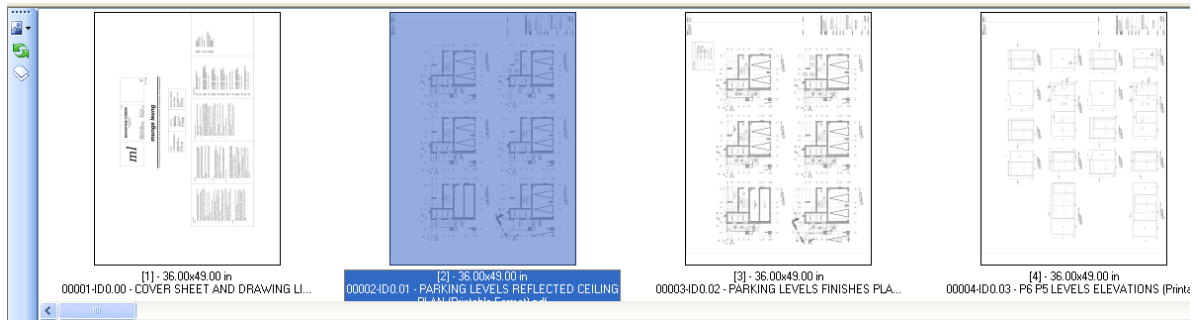


The full screen viewer is not recommended for frequent and ongoing job setup, but rather to check some content or the processed result on some files, once in a while because it required additional processing depending on your configuration. There is no alignment option in the fullscreen viewer.

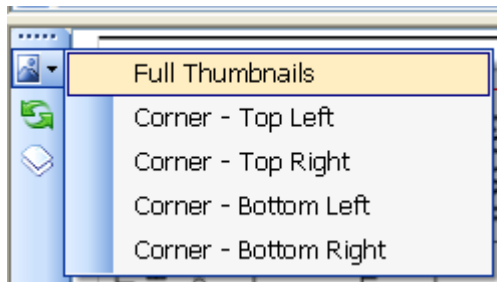
The on-screen editor viewer can be configured to give good enough resolution and will suit almost all preview needs much faster.

Thumbnail Viewer:

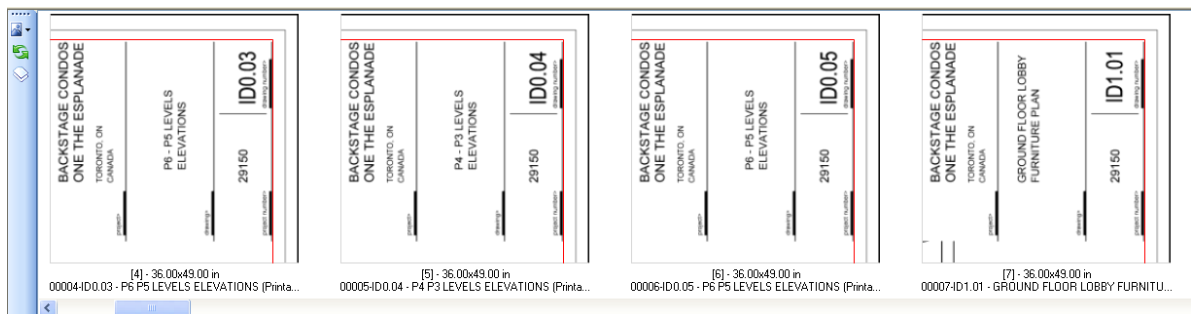
The bottom part of the screen is a scrollable thumbnail viewer. You can preview multiple files at once for rotation or general content quickly. Selecting one or more files and right click will rotate them in steps of 90 degrees clockwise.



There are 4 special thumbnail modes that show a corner only for precise edge alignment. See config section 02 for details and size of thumbnails. The position of the red guideline is a configurable setting as well.

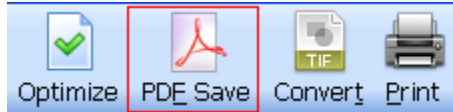


The thumbnail area also shows sheet sequence number in job, filename, page number for multipage files and page size.

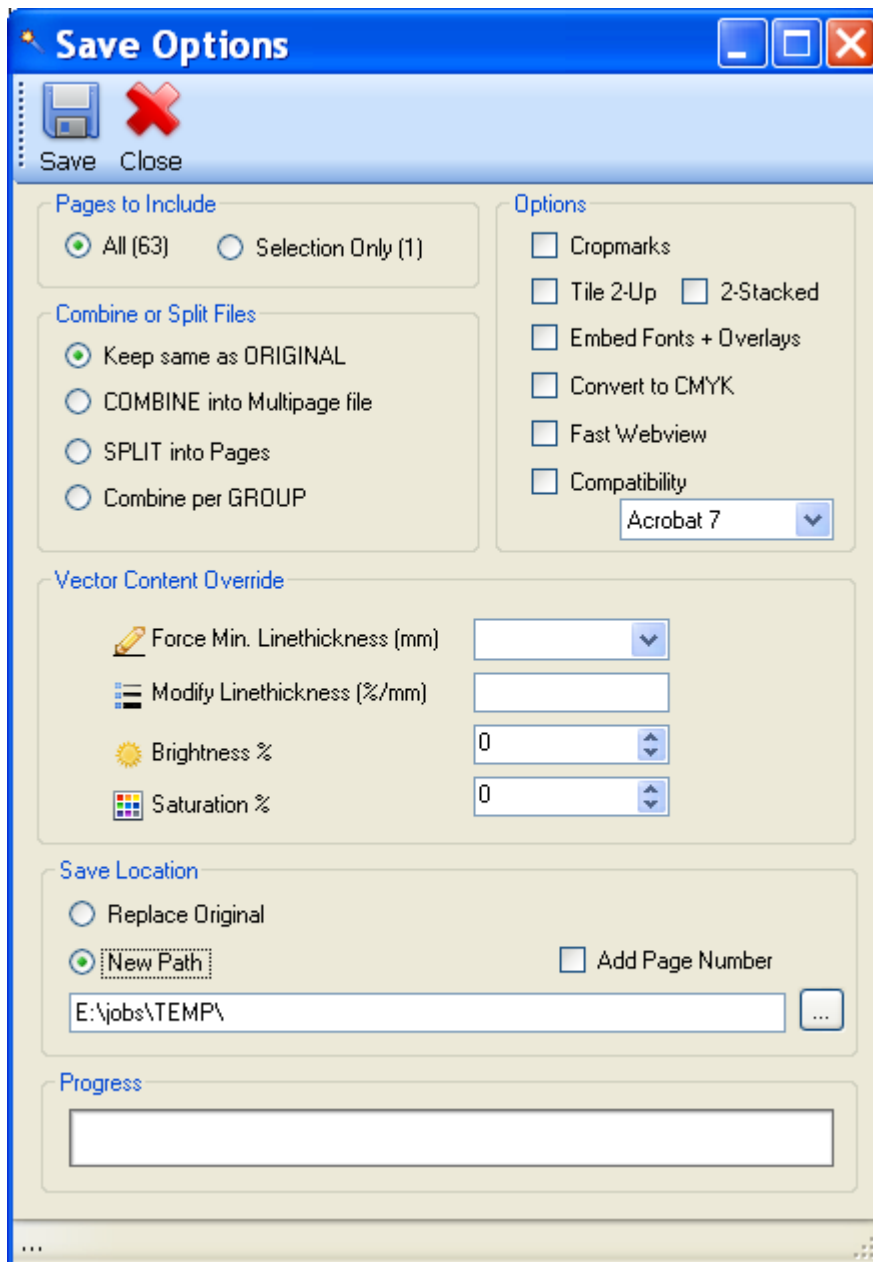


Exporting to PDF files:

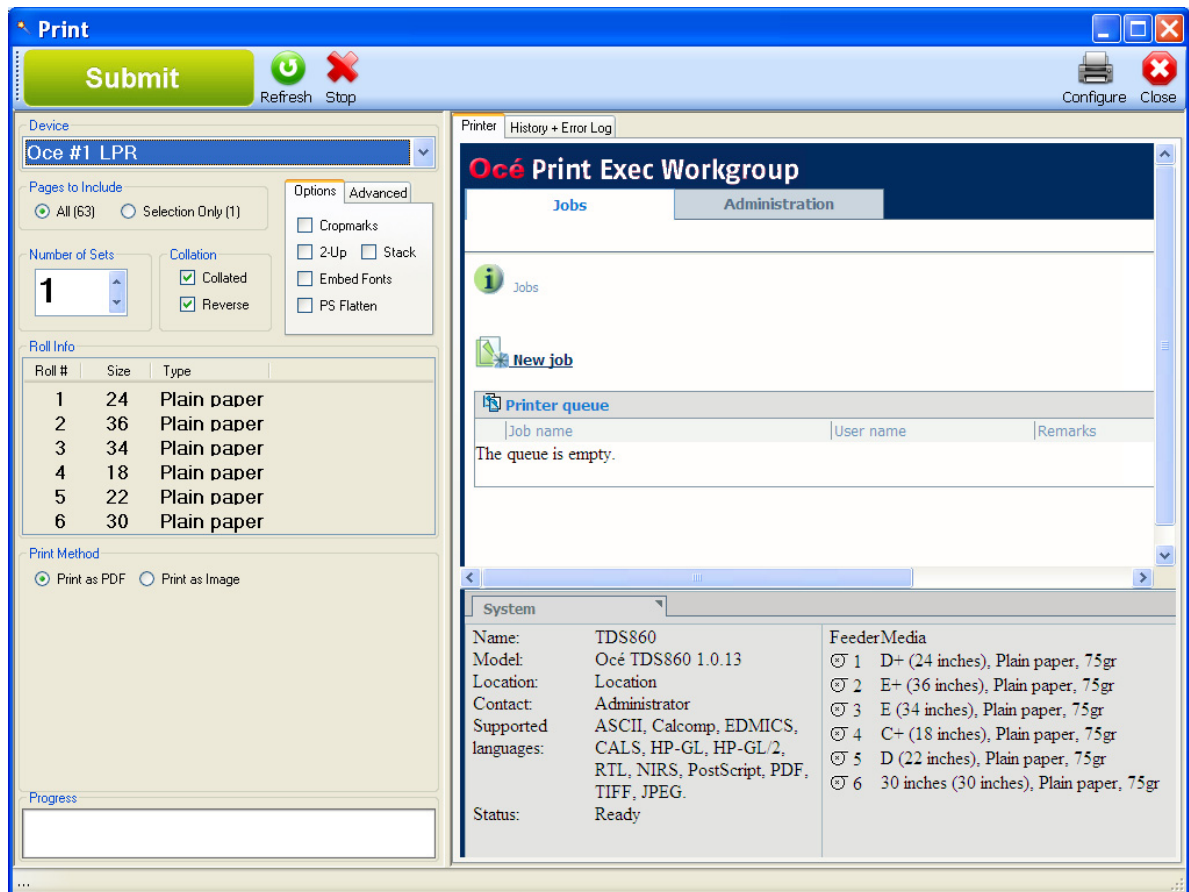
Click on “PDF Save” to open the Save options screen.



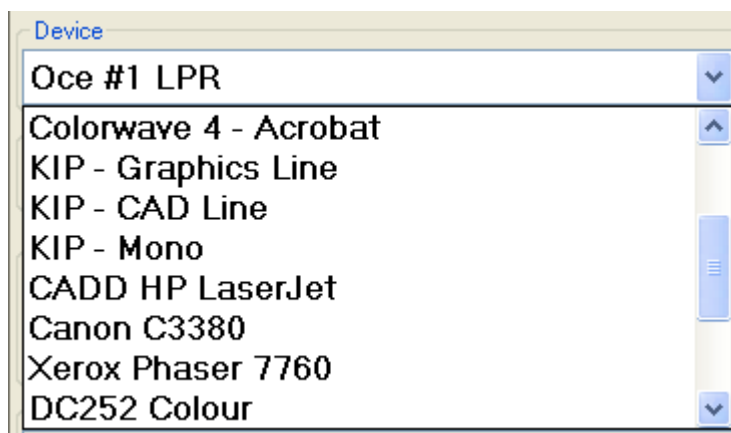
Select desired output options. You can “Print” to a “Hotfolder” that is monitored by 3rd party software as well. This screen will commit all modifications done in the editor to PDF file(s).



Printing to a Device:



Clicking on “Print” will open the Print Dialog. Depending on your Printer Configuration you may have one or more devices. Selecting a device will refresh its device specific interface and show information about current status and media.

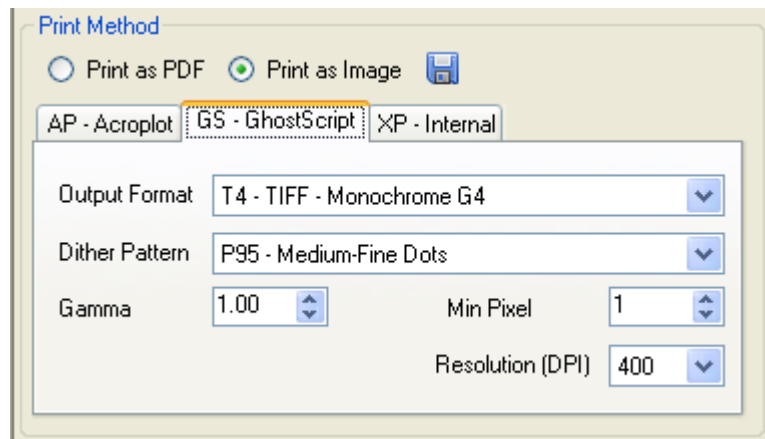


Device specific defaults should be configured on its own interface, on the right panel or on the separate controller of the device. If the device is configured with “Acrobat_Options” protocol, you will get a print driver dialog where all device specific options can be selected after you click “Submit”

Printing as PDF or Printing as Image:

By default “**Print as PDF**” should be selected. It is far more efficient and preserves the original PDF on the device. If the device is not PDF capable or has printing problems in PDF mode, Print as Image can be selected. Print as Image is primarily designed for Largeformat devices.

“**Print as Image**” does a similar rasterization what the rip/controller would have done internally. Please review **Chapter 7** for detailed information. Experiment with different output options depending on your device to get a feeling for speed, dithering patterns and quality.



If you are working in a “Queue Based” workflow; It is possible to have a powerful setup PC do the processing to image in advance and queue required print ready images with the job. A separate print station can open and submit using those files without additional processing time.

Also in a Queue Based workflow, if station “A” prints as image, the same files will be available to any other station automatically as long as both stations are working from the same queue. The printable image files will be invalidated and recreated automatically when required if the operator changes job properties like zoom, orientation, alignment etc. and tries to print as image.

In both print modes collation, 2up, 4up, font and overlay embedding, cropmarks options are available. Other advanced print options include vector content linethickness override, brightness and colour saturation controls. Please review training videos for more details. For Print as Image options resolution, dither pattern, brightness and minimum pixel are available. All of these settings give the user a lot of flexibility to improve and adjust print quality but require an experienced operator and understanding what they are designed to do.

“There are a thousand ways to print a drawing wrong and only 1 way to print it right...”

Configuration of User Settings

This section explains configuration options that users can adjust to customize the application to suit their needs and type of jobs. Each option has detailed explanation at the bottom of the property grid.

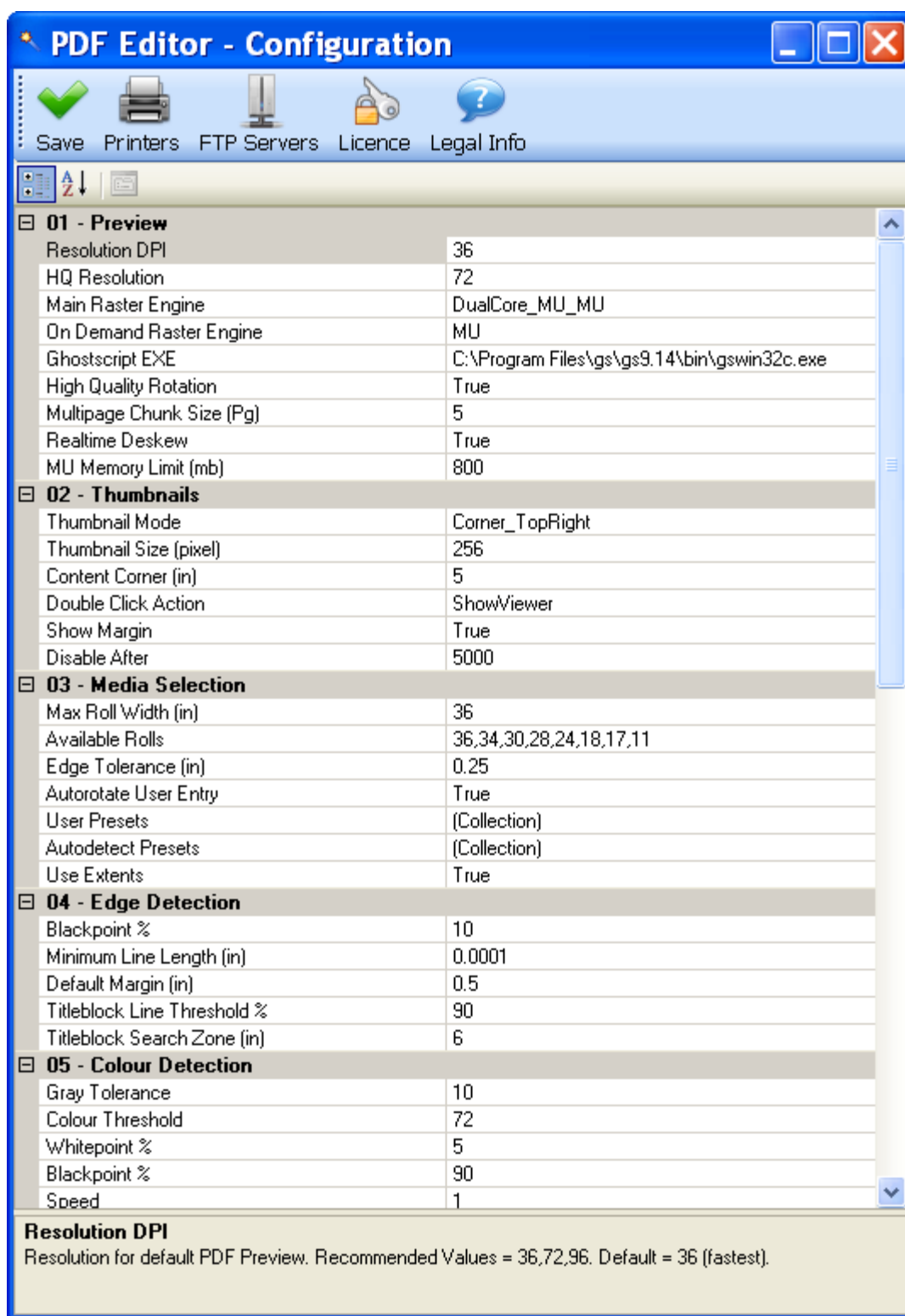
01 - Preview:

All values in this section affect preview image generation only. The recommended resolution for largeformat is 36/72. For smallformat it is 72/96 (default/HQ dpi). There is no need for operators to see high resolution preview images when preparing a job for print or rotating and aligning pages. Preview resolution quality will never affect output (export) or print quality. Exporting and Printing is always done in native (original) full vector resolution. The MU preview engine is default and recommended for most cases. It is the fastest. If you encounter preview problems, like blank pages or preview not matching original file you can change the preview engine to XP or GS or LN or GM. They are recommended to be used in that order, based on performance.

02 – Thumbnails:

This section controls the appearance of the thumbnails at the bottom of the main editor. Thumbnails can show the full image – for orientation or overall content check, or just any one of the 4 corners for precise alignment or sequence check. Generating thumbnail images takes CPU power. If the station is a low end PC used for Printing only and jobs are setup on another PC you may want to turn off thumbnails completely. Thumbnail images are cached in memory. For very large jobs the PC may not have enough RAM and the system will destabilize if too much load is put on memory. To avoid this, PDF Magick will turn off thumbnails automatically after a cut off limit that can be configured. With thumbnails off, there is no reasonable limit to the size of the jobs that can be handled.

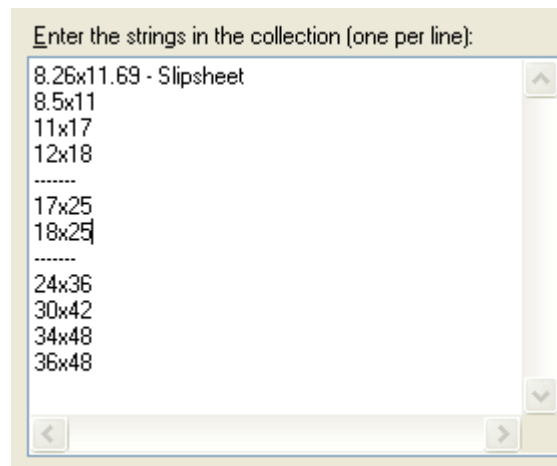
< continued >



Screenshot - User Configuration - Part 1

03 – Media Selection:

The Editor has multiple “Autoselect” options. Its behaviour is controlled in this section. You can define the roll widths available at the shop in imperial or metric – “Autodetect by Edge” will use this list. You can also define common and known papersizes and they will be available as a quick selection in the editor. Another option is to configure “Autodetect Presets” – “Autodetect by Sheet” command will use this list. This is very useful for breaking up a job with many sheet sizes into chunks of known sheet sizes and can be used for smallformat and largeformat. “Autorotate = True” setting allows you to ensure that the long edge of the media is always aligned with the long edge of the image; it can be turned off for special cases. “Use Extents = True” is useful to “Crop white margins” or use the detected extents as the base for media selection. Extent detection command must be run for this to activate. Use “mm” for metric values. Example “210x297mm”



Sample values for "User Presets"

04 – Edge Detection:

The editor can detect edge lines and align the image on media based on those results to any edge or corner. This avoids having to manually align sheets when each sheet has a different margin. This section controls the values for the detection. Based on the value of “Minimum line length” you can make the software detect a single dot as an “edge” (use a very low value like 0.0001) or a solid titleblock line with a minimum defined length (use 4 to 6”). This way you can make the software ignore possible cropmarks or file name and date stamps and target the main edge line for alignment.

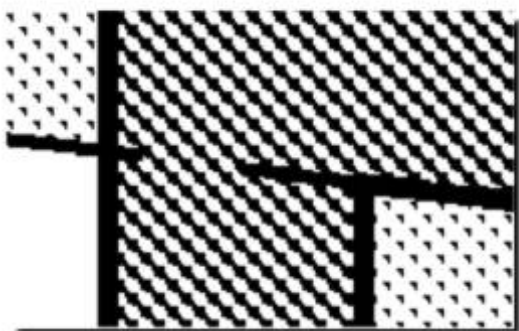
05 – Colour Detection:

The editor can separate BW and Colour pages and detect % coverage. It can automatically tag colour pages and bulk select them. Detection can be tuned to be very sensitive and detect a single dot of colour (use a low value for “Colour Threshold”) or tuned less sensitive and look for large areas of colour only (use a high value for “Colour Threshold”). Please review the description of each value for further details and experiment with the results.

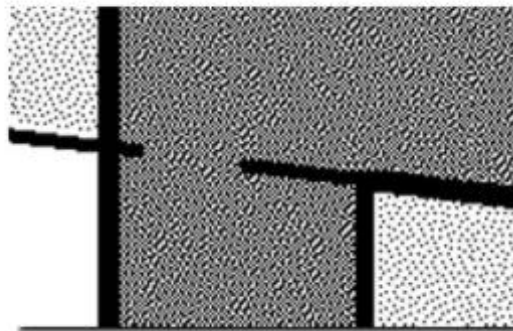
06 – PDF to TIF Processor:

There is no method or software that works 100% correct on all files in all cases. This section controls the optional PDF to Image renderer if native PDF processing or printing has problems. Also if your largeformat device does to have native PDF capability printing as image is an option. There are 3 separate image processors, in case one has problems with a particular file. Ghostscript Acroplot Repro and a special internal one codenamed “XP”. Ghostscript is constantly updated by its developers and can be downloaded from <http://www.ghostscript.com/download/>. Acroplot Repro is licenced software and can be obtained from <http://www.cadzation.com/downloads.htm>. There is no easy answer to the question “Which one is best.” Each processor has its strengths and weaknesses depending on the file to be processed. There is no single processor that works 100% correct in all cases on all files; that’s why PDF Magick has more then one integrated. Users are encouraged to test out the different dithering patterns and options available and review Chapter 7.

Ghostscript® at 300 DPI



AcroPlot Repro™ at 300 DPI



07 – Adobe Acrobat:

This is the link to the executable for Adobe Acrobat. Either the free Reader or any professional version will be fine. This is used when “Open in Editor” is selected or when printing with one of the “Acrobat” protocols.

08 – PDF Output:

This section controls cropmark style, a full box or corner marks only. In the future other PDF output related options will be added here.

09 – Queue:

This is the location of Interbranch path for operator reference used in the “Send” screen. Read only. It can be configured from the application configuration file as explained in chapter 1. Setting name **Queue_IB_Root.**

The image shows a 'PDF Editor - Configuration' window. It has a blue title bar with standard window controls. Below the title bar is a toolbar with icons for Save, Printers, FTP Servers, Licence, and Legal Info. The main area is divided into sections, each with a expandable icon (a square with a minus sign). The sections are: 06 - PDF to TIF Processor, 07 - Adobe Acrobat, 08 - PDF Output, 09 - Queue, 10 - Job Editor, 11 - Slip Sheets, 12 - Guidelines, 13 - Duplex Setup, 14 - Job Summary, and 15 - Version. Each section contains a list of configuration items with their current values. At the bottom, there is a 'Resolution DPI' section with a note about recommended values.

06 - PDF to TIF Processor	
Acroplot EXE	C:\Program Files\Acroplot Repro\AcroPlotRepro.exe
Optimize for CPU	DualCore
Simplified Dither Options	False
Multipage Chunk Size (Pg)	5
Preprocessing Mode	True

07 - Adobe Acrobat	
Acrobat or Reader EXE	C:\Program Files\Adobe\Acrobat 7.0\Acrobat\Acrob

08 - PDF Output	
Cropmark Style	Marks

09 - Queue	
Interbranch Root	\\vrt-svr-flp-02\shared\I - Interbranch - I\

10 - Job Editor	
Insert Key Action	Insert_Blank
Units	Imperial
Log Destination	ToFile
Double Click File Action	ShowHQPreview

11 - Slip Sheets	
Slip Sheet Font Size	24
AutoInsert Sheet Size (in)	8.26x11.69

12 - Guidelines	
Show Guidelines	False
Default Positions	(Collection)
Default Colour	Red
Default Linestyle	Dash

13 - Duplex Setup	
Section Pattern	SECTION.\d+\s\d+\s\d+(SECTION.\d+\s\d+(SECTI
Keywords for Fronts	(Collection)
Keyword Search Lines	6
Min Line Count	6
Preferred Paging - No Section	DoubleSided
Preferred Paging - No Text	DoubleSided

14 - Job Summary	
Smallformat Cut Off	11x17
Area Calculation Rounding	RoundUp

15 - Version	
Version	40527
Update Path	\\vrt-svr-flp-02\shared\PDF Magick 4.0\

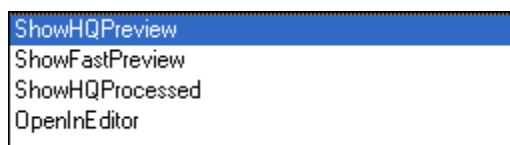
Resolution DPI

Resolution for default PDF Preview. Recommended Values = 36,72,96. Default = 36 (fastest).

Screenshot - User Configuration - Part 2

10 – Job Editor:

This section allows selection between metric or imperial units and other overall editor related options. Error messages during processing can be shown immediately on screen, which will stall the processing until the operator acknowledges the message or can be logged to file and the log file will be displayed at the end of the process. “Double Click File Action” has multiple options.



ShowHQPreview : Render original PDF using HQ Resolution DPI and display. Once a HQ version is displayed it is cached and if you double click the same file again, the previously generated version is used to speed up display.

ShowFastPreview : Use existing preview image and display.

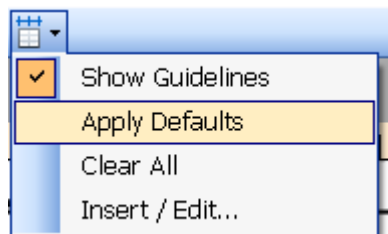
ShowHQProcessed : Apply all processing modification to a temporary PDF, as if printed or exported, then render using HQ resolution DPI and display. This will show deskew and overlays.

11 – Slip Sheets:

The editor can insert “slipsheets” automatically between groups to separate them easier. In this section you can configure the properties of those slipsheets. The font size is used for the “Group” name on the autoinserted slipsheet. For example if you drop a lot of files from multiple directories you can have a slip sheet for each directory with its name automatically inserted.

12 – Guidelines:

The main viewer will show rulers when in zoom mode. Double clicking on the rulers itself will create a “Guideline” at that spot. You can move the line with the mouse as well. Guidelines created within the viewer will be attached to the active job and cleared when a new job is started. If you want global guidelines for all jobs, for example a 1” margin on the left to see hole punch positions you can configure them in this section and then apply them to any job with “Apply Defaults” in the viewer.

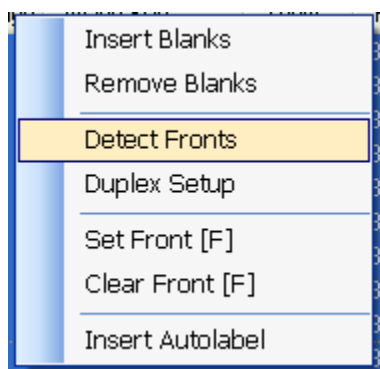


13 – Duplex Setup:

The Editor has a “Detect Fronts” option which will attempt to mark “Front” pages like new section beginnings in a multipage standard structured specification document. This section allows customization beside the built in rules. For example if certain “Keywords” are defined and present in the page content, they will be automatically marked as “Front”. Not every PDF is compatible with the internal reader. For documents that are compatible, a 2,000 page document can be setup for duplex printing and blanks inserted in a few seconds. Scanned documents must be OCR’ed separately and are not compatible with duplex detection directly in scanned format. Some keyword suggestions are “Table of Contents”, “Appendix”. If you need a reference for the “regular expression” used for the “Section Pattern” field please see :

<http://www.codeproject.com/Articles/9099/The-Minute-Regex-Tutorial>

The default value of `SECTION.\d+\s\d+\s\d+|SECTION.\d+\s\d+|SECTION.\d+` will match sections formatted similar to SECTION 12 00 00 or SECTION 1200 00 or SECTION 1200.



14 – Job Summary:

This section controls preferences for Job Summary creation. Separation size for Smallformat vs. Largeformat can be defined. Also area calculation method for Largeformat can be defined as “Round Up” or “Fractional”

FAQ & Troubleshooting

STUTTERING DURING SUBMISSION ON PC:

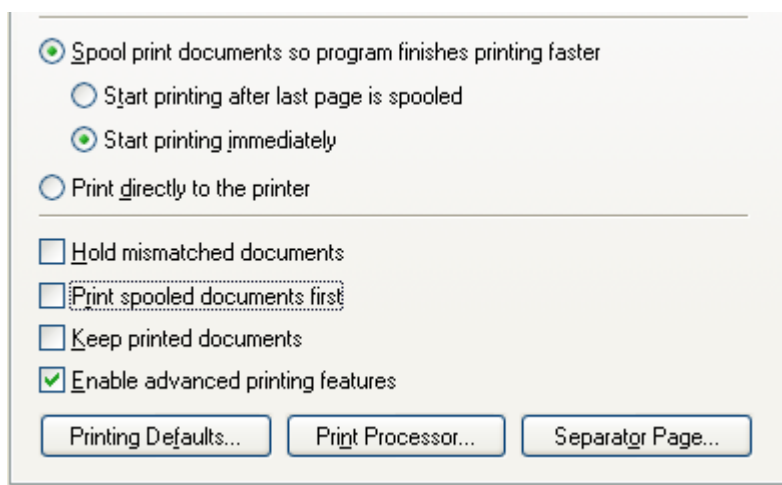
If you experience “stuttering” during submission, where the print submission progress bar stops for 1~4 minutes approximately **every 11 sheets** and submission protocol is **LPR** please see <http://support.microsoft.com/kb/2003449> and create the required registry key. Ensure that the key is of BINARY type for windows Vista, 7 and 8 and DWORD type for Windows XP. You can also ask your PDF Magick support person for a registry file that will adjust this for you automatically.

STUTTERING DURING PRINTING ON DEVICE:

If your largeformat device is stopping unreasonable long between pages and you have been printing in PDF mode it may be possible that the onboard rip is not powerful or efficient enough for the complex vector data in the PDF. Some PDF's with complex transparency layers can cause this. The easiest workaround is to “Print as Image”. You can “Test” which Image processor works best. There is no single processor that is the best quality and fastest in all cases. Each of the processors have their strengths and weaknesses. You will need a powerful PC for difficult files.

STARTING TO PRINT TAKES TOO LONG:

On a Largeformat device the first page should start printing around 15~30 seconds after submission; depending on your device. If it takes multiple minutes and gets longer with larger jobs ensure that the device is configured with “LPR” protocol and page submission method is “Singlepage” and that “Print while processing” is “True”. If the device has to be configured with one of the “Acrobat” protocols in multipage mode because of capabilities or connectivity, initial startup time will take a bit longer. Check device print driver advanced properties that “Start printing immediately” is selected. Some devices cannot print while receiving and processing more information at the same time. This is a restriction of the device and not PDF Magick.



CAN I WORK WITH IMAGE FILES IN PDF MAGICK:

Yes. Besides PDF format common image formats are supported as well. BW or Colour TIF, JPG, BMP, PNG files can be dropped into the editor just like PDF files. The editor will convert them to PDF internally. Use caution when reducing monochrome TIF files that have 1 pixel or very thin lines. When files are added from subdirectories only PDF files will be imported.

CAN I WORK WITH HPGL/2 PLOTFILES IN PDF MAGICK:

Not directly. PLT files are not supported directly by the editor. The easiest method to work with HPGL/2 files is to use a 3rd party tool and convert PLT to PDF. Please try Viewcompanion at <http://www.softwarecompanions.com/> It supports batch conversion. There are other tools that can do this conversion as well.

CAN I WORK WITH AUTODESK DWF FILES IN PDF MAGICK:

Yes. PDF Magick has a Vector based DWF to PDF Converter which preserves the quality of the original vector DWF. Once configured you can drag & drop DWF & DWFX files into the editor just like PDF files and they will be converted to PDF for you. There are other tools that can do this conversion as well. Please try Viewcompanion at <http://www.softwarecompanions.com/>.

THE PREVIEW SHOWS BLANK PAGES:

Check the actual PDF, if it is fine change your preview Rasterizer Engine. If it was MU, try XP, GS, LN, GM in that order. If that does not help open the file in Acrobat, re-save it with no changes and try again or "Print" to a new PDF file from Acrobat. This will rebuild the PDF stream in case there is something incompatible with the preview engine. The program will automatically switch back to "MU" when it is restarted because it is the fastest.

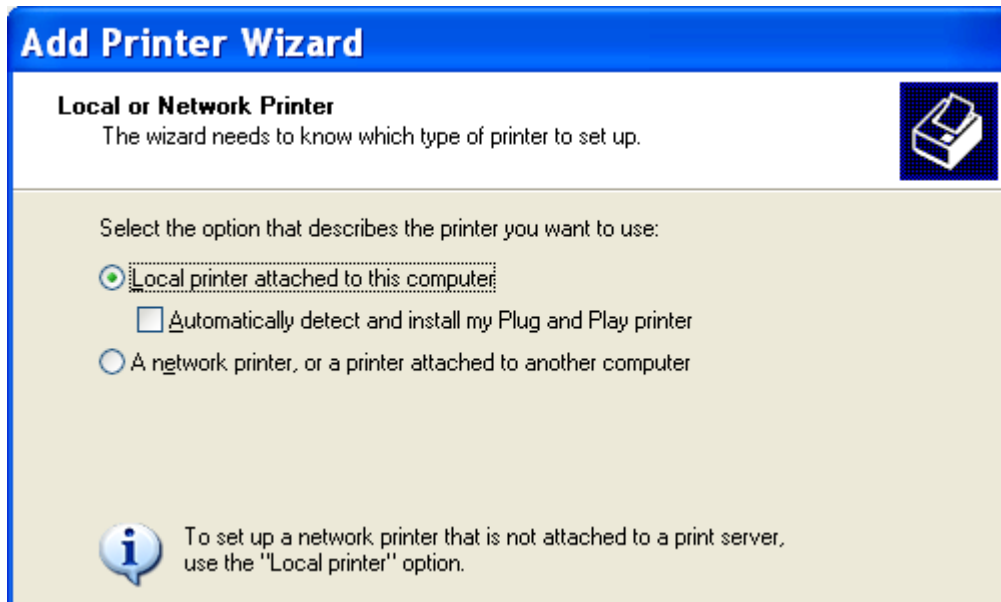
HOW CAN I ROTATE CONTENT AND MEDIA SEPERATELY:

By default autorotate in config is enabled and the editor will align the long edge of the content to the long edge of the sheet. You can turn this behaviour off in config section 03 Media Selection. If you need free rotation rarely it is better to leave the config as is and do the following : Rotate your content first; do not worry about papersize. Then select Papersize > Custom > and enter 17x11! Or 11x17! Depending on your desired size. Note the ! sign at the end. The first number is the width. Autorotate will be disabled when there is a ! sign at the end.

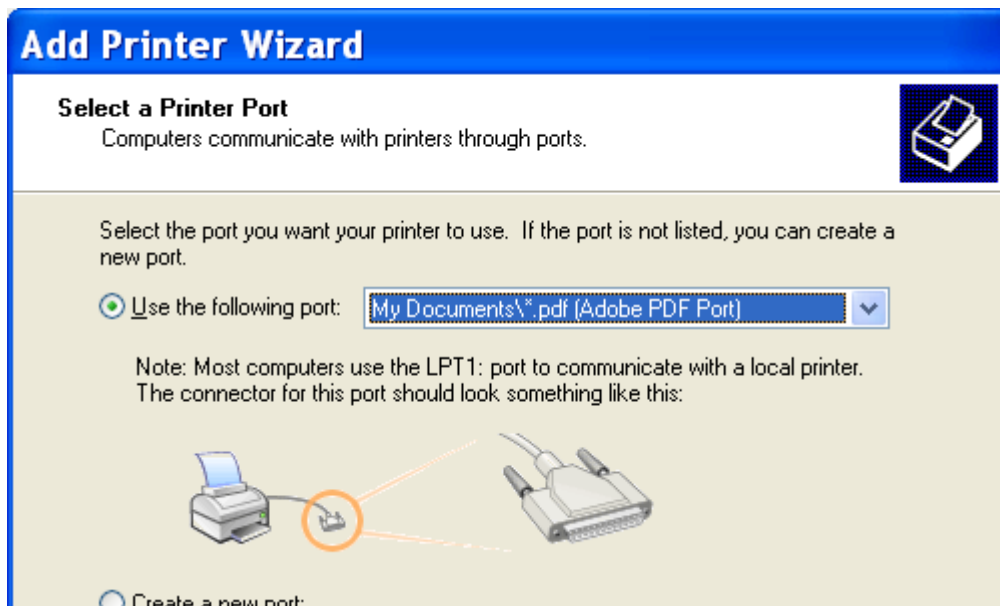
INSTALLING AN ALTERNATIVE DWF TO PDF CONVERTER:

If you do not want to modify the output path of your existing Adobe PDF driver you can install an alternative driver. The steps are explained below.

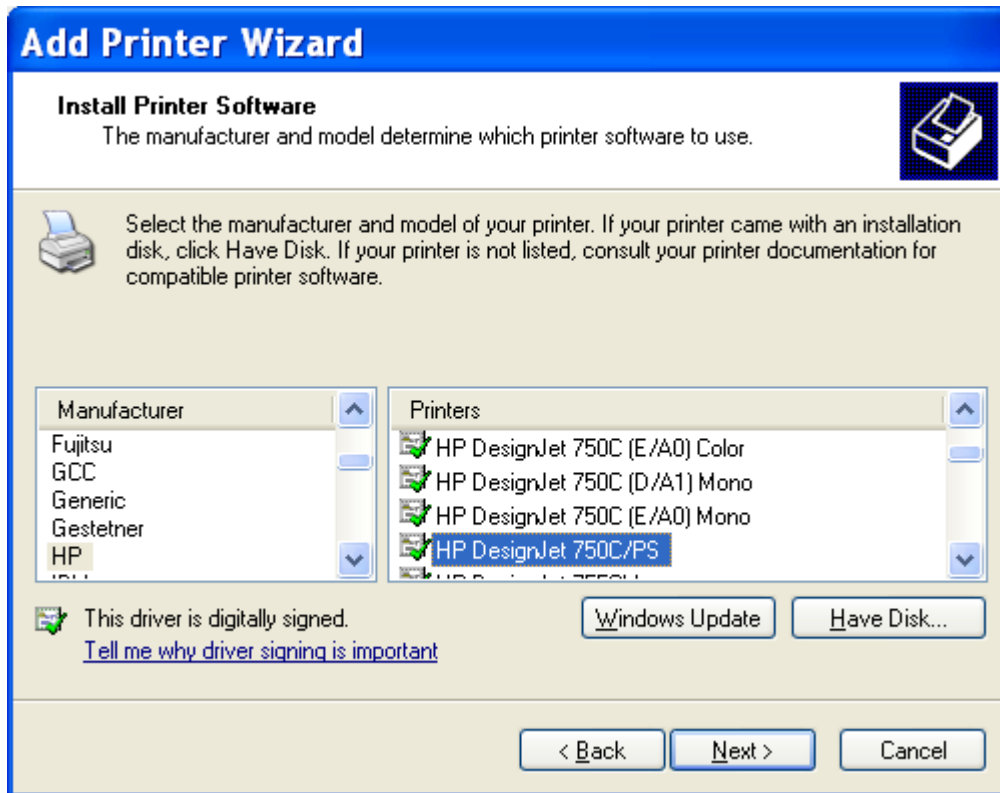
Start with “Add a New Printer”



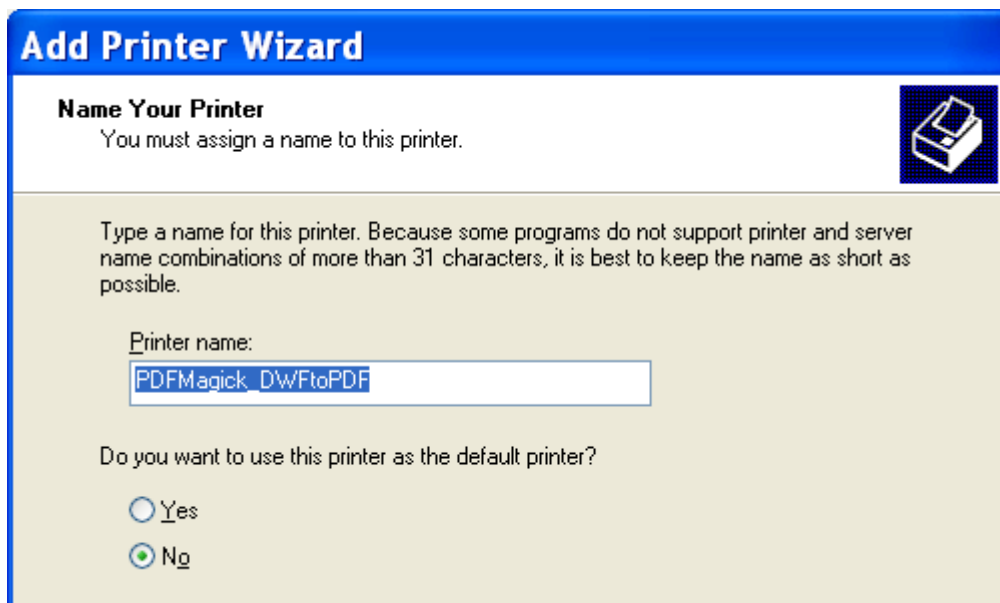
Point it to the existing Adobe PDF port. If you do not have this port you must install Adobe Acrobat PDF driver first that comes with the full version of Adobe Acrobat.



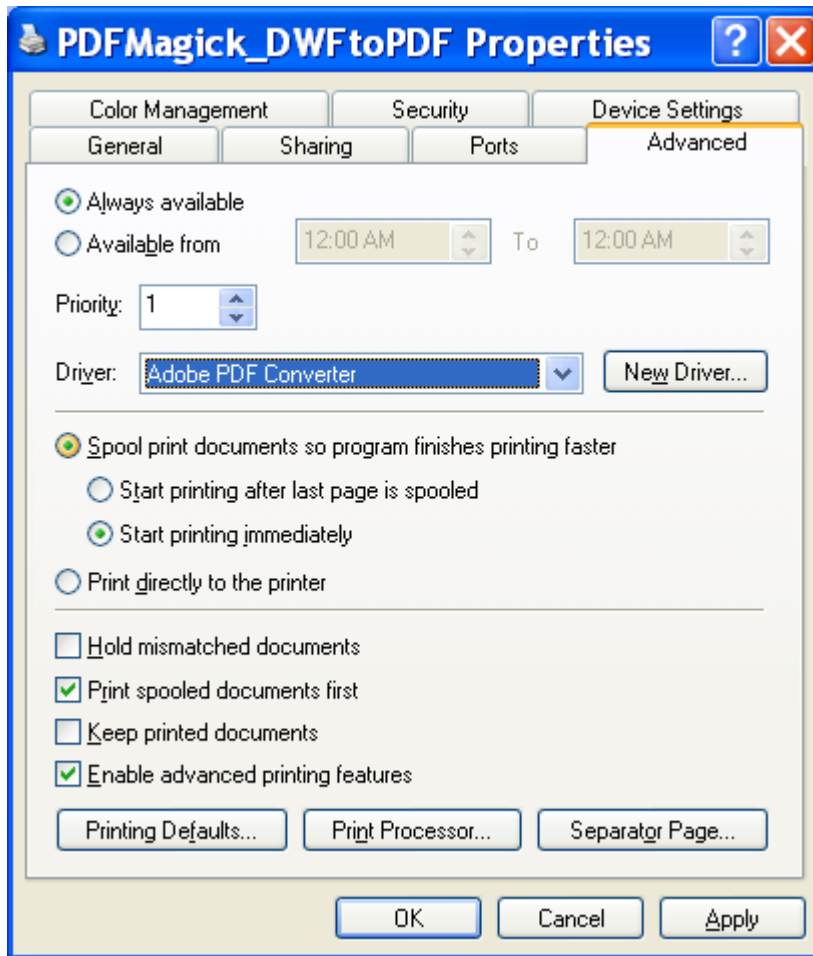
Select Manufacturer = “HP” and Printer = “HP DesignJet 750C/PS” this is intended as generic postscript driver used for conversion only.



Give it a suitable name and finish the driver installation

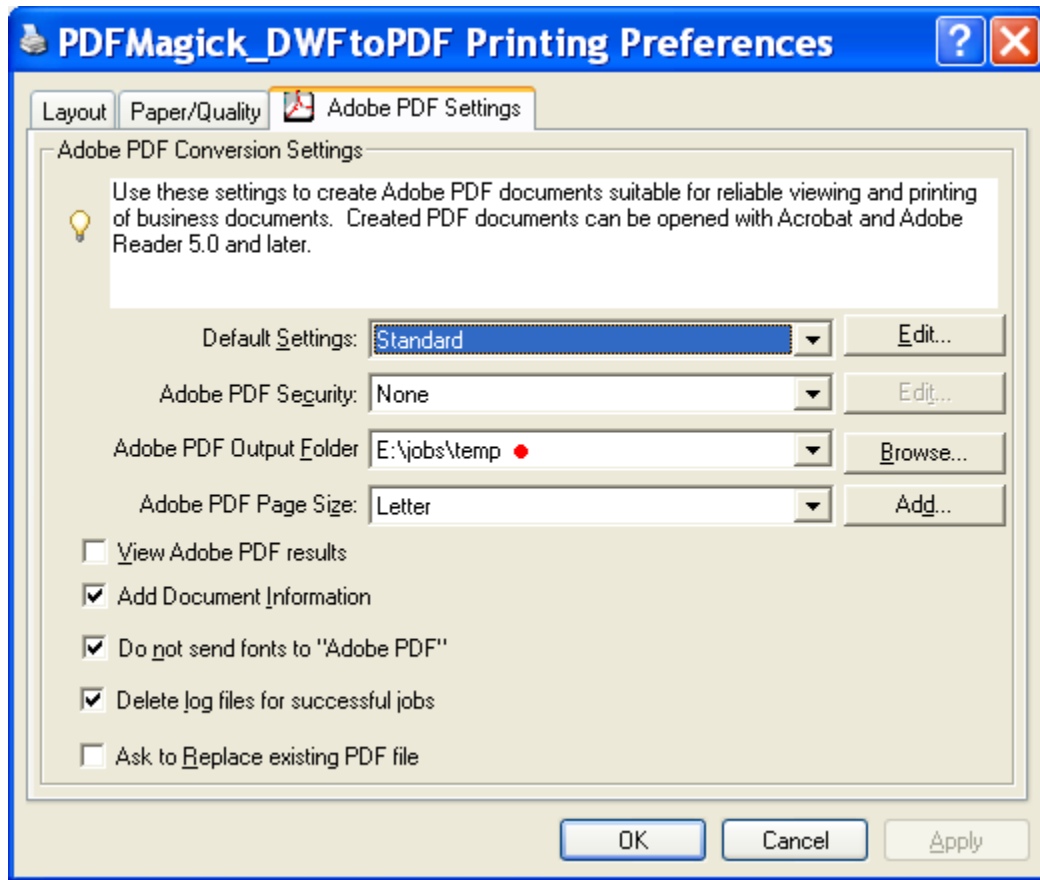


Adjust the properties. “Driver” is critical and must be changed to “**Adobe PDF Converter**”



Enter the Printing Preferences and adjust “**output folder**” to a temporary folder.

Also uncheck “View Adobe PDF results”



Finally adjust PDF Magick configuration section to match the driver.

15 - DWF Importer		
DesignReview EXE	C:\Program Files\Autodesk\Autodesk Design Review 2013\DesignReview.exe	
PDF Driver Name	PDFMagick_DWFtoPDF	
PDF Output Folder	E:\jobs\temp	

 **Note :** If your original DWF is oversize; with pages larger then 36x48” or sheet sizes that are very irregular you need to configure a “close enough” and preferable somewhat larger PDF page size in the Printing Preferences of the driver. Otherwise your content might get cut off.

TIPS FOR DEALING WITH TROUBLE PDF FILES:

Some PDF files might give trouble during preview, setup, exporting or printing in form of stalling, crashing or incorrect output. The PDF format is still evolving and there are many non-Adobe drivers and software that create and modify PDF files. There is no single solution that solves all PDF problems. Nothing beats an experienced operator that understands the source of the problem.

Below are some general tips what to try if you encounter problems:

- Open file in Acrobat; check nothing wrong; resave with no changes in Acrobat. This fixes some PDF stream and (xref) problems if the PDF was created by non-Adobe software.
- "Optimize" file in PDF Magick with "Embed" = ON. This rebuilds the PDF stream completely and if the problem was on the software/PDF Magick side fixes most problems.
- You can "Optimize" to a lower PDF version as well. Note that "Optimize" and "PDF Save" use a different code path & sequence of events. If the PDF breaks in PDF Magick during "Save". Optimize should be attempted as it works with the original PDF at a lower level using another processing engine.
- You can remove transparencies completely if "PS Flatten" = ON; but it is a dangerous operation for some files; it might compromise some content & sequence of objects. The output should be checked carefully. PDF Magick will give a warning message if it detects a severe problem during flattening.
- You can rasterize to TIF/JPG - which will fully flatten the file using any of the PDF Magick renderers and then import the image back into PDF Magick; it will convert it back to PDF; but only with a "jacket" the content is now raster and will work well on any rip or software. But this can compromise or maybe lose some quality if incorrect rendering resolution or settings are chosen.
- You can open in Acrobat and "PRINT" to the Acrobat PDF Driver; with custom settings; this rebuilds the PDF stream completely as well.

Printing Modes – PDF vs Image

Read this section if you want an in depth understanding what the difference is between the main 2 print modes and why there are 3 different image processing engines each with different settings. This section is primarily for largeformat devices.

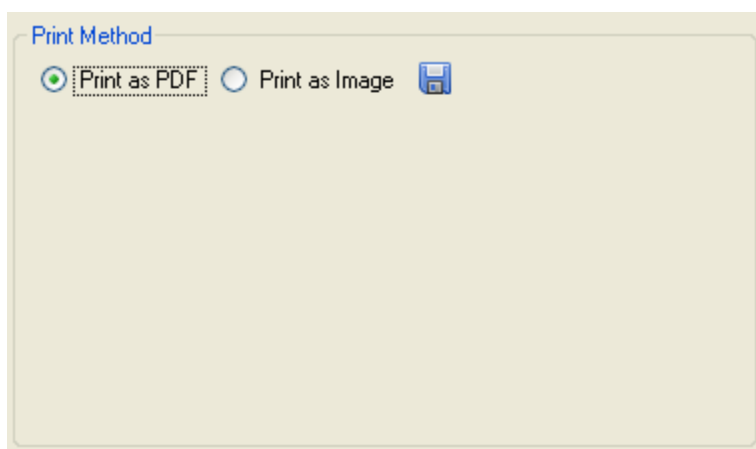
When printing in PDF mode all the “hard” work of turning the file into a printable image is done by the onboard RIP/Controller of the device and its internal settings. The PC has very little work to do. The device must be capable of native PDF processing.

When printing in Image mode all processing is done on the PC, with the help of a processing engine that converts the PDF to an Image format. The device does not need to be PDF capable and has very little work to do.

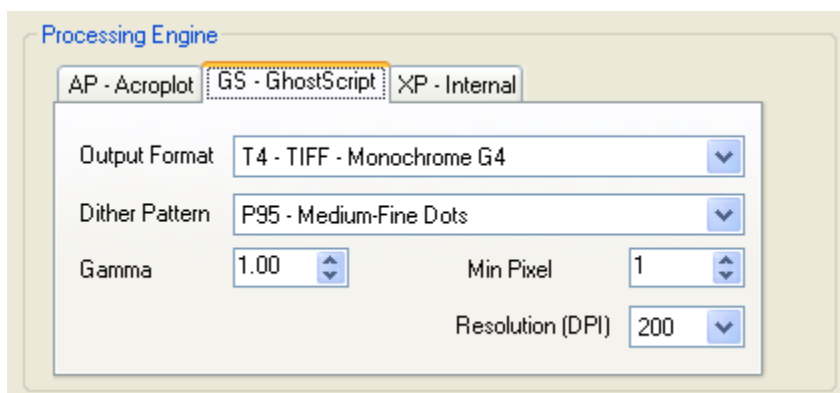
The above statement summarized the core difference between the 2 print modes. Having a PDF capable device and printing in PDF mode is the ultimate experience and what PDF Magick is designed for. The PC will be free for the next job very quickly and there is less to worry about print quality because the device RIP/firmware is tuned by its manufacturer so well for that particular device. Printing in PDF mode also means that you are immune to quality loss or pixellation when reducing or enlarging files as long as the input file is vector based.

However there is no device, rip, software or processing method that works 100% correct in all cases and on all files. You may be forced to print as image if the device fails to print as PDF properly. Also some devices may not have native PDF capability at all or it may be disabled due to manufacturer's additional license costs. That's why PDF Magick has a very advanced “Print as Image” option.

PDF Magick will process files and clear the PC for the next job 30 to 100 times faster in PDF mode compared to TIF/Image based other software. In Image mode PDF Magick will be 2 to 5 times faster then other software if a dual or quad core CPU is used.



“Print as PDF” - There are no settings for the user in the Print submission options because all settings are controlled by the RIP/Controller. You can adjust device defaults & preferences using the interface of the device itself.



“Print as Image” – There are 3 different processing engines, each with different settings. Output quality and processing speed will be impacted by the choices made here.

Image Processing Options:

There are 3 different Image Processing Engines because the fact that no software or processing method works 100% on all files in all cases applies here as well. Some PDF files behave better, due to quality or speed, in one processing engine than the other. No other software gives you this many processing options and it is important to understand the strengths and weaknesses of each processing method. No software can replace an experienced and trained operator that knows how to use the tools available.

AP - ACROPLOT:

Dither Pattern : ED – Enhanced Dithering – This is a very reliable and well balanced pattern for files with fine gray lines as well as images. Areas with solid shades of gray will have a fine dot pattern and might show device service & cleaning problems more compared to a grainier pattern. Possible problems are embedded OLE objects with wipeout and internal rotation. Dark gray area's or solid black areas may appear on some files but it is rare. Some files with complex transparency objects may take a long time to process compared to other engines because of the special "Line Merge" features used.

Recommended for linework and basic images.

Dither Pattern : C36 – Clustered Dithering 36 shades – This is a coarser pattern than ED. It has more levels of gray.

Recommended for large solid area screens at different tones or large images with wide ranges of gray.

Patterns with higher C values are coarser and give more levels of gray tones. They are suitable for posters and graphics with lots of levels of gray but a poor choice for fine gray linework because pixels will drop.

Acroplot is separately licensed software and can be obtained from its vendor at https://www.cadzation.com/acroplot_repro/acroplot_repro.htm.

GS - GHOSTSCRIPT:

In many cases the GS pattern are extremely fast, possibly the fastest of all 3 rendering engines and very reliable. GS patterns are coarser than AP patterns.

Dither Pattern : P95 – Default if AP fails due to wipeout or takes too long or AP is unavailable. GS-P95 pattern is more coarse than AP-ED. This pattern will show overexposure or gray level "jumps" on images because it has limited levels of gray.

Recommended for linework.

Dither Pattern : P75

Recommended for a balance of linework and solid gray tone screenings.

Dither Pattern : P60 – Coarser dots, not suitable for fine gray lines – it will look pixellated.

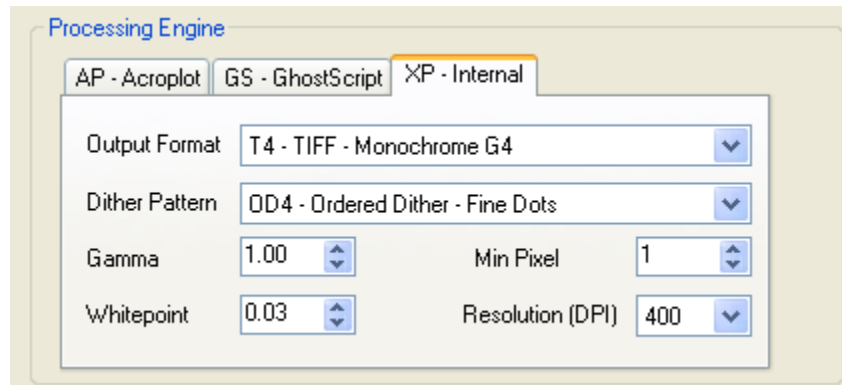
Recommended for Posters or Images that require a wide range of gray tones or to hide device service & cleaning problems.

Lower P values are coarser and give more levels of gray tones. They are suitable for posters and graphics with lots of levels of gray but a poor choice for fine gray linework because pixels will drop.

XP - INTERNAL:

XP Engine is designed for files that have very complex transparency objects. It can outperform AP and GS on a margin of 5 to 1 in some cases. The XP engine also has more configuration options and pattern types to give the user more options to control the output.

If logo's or embedded small images show a “ghost” or light gray background, the “whitepoint” option of XP can be set to treat them as white which will eliminate the unwanted background.



Dither Pattern : OD – Ordered Dither. Dots are spaced in order, along the horizontal and vertical.

OD4 – Recommended for fine linework and basic gray screening.

Dither Pattern : CD – Clustered Dither. Dots are spaced at an angle.

CD4 – Recommended for fine linework and basic gray screenings.

CD6 – Recommended for images and levels of gray.

Dither Pattern : SD – Stochastic Dither. Dots are spaced with an error diffusion pattern

The higher the dither code number the coarser the dots; meaning more levels of gray but possible pixellation on fine lines.

COMMON OPTIONS :

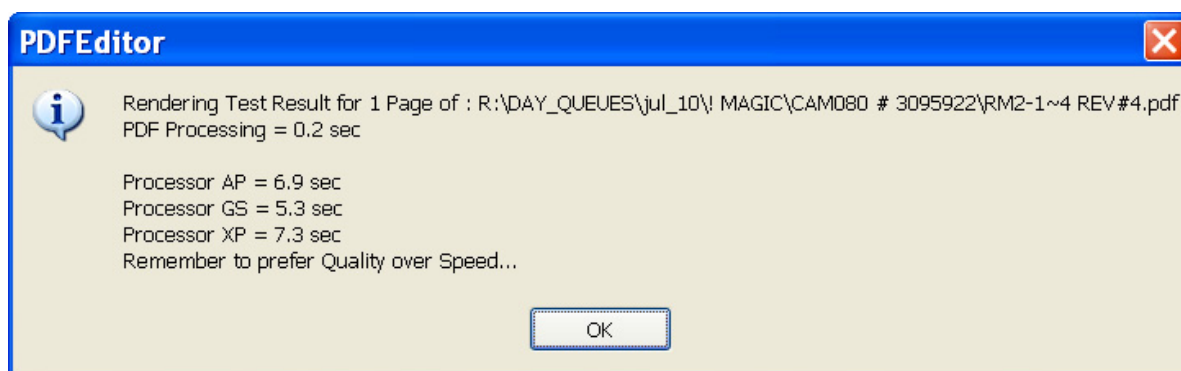
For all patterns there is a **brightness** and **resolution** setting as well. The resolution should be chosen to match the print head of the device, the user is encouraged to test different options and use what works best on his device. A separate **minimum pixel** control can be set to 1, 2, or 3 to ensure very thin lines in the PDF do not disappear when printing.

Dither patterns apply to T4 – Monochrome TIF output format only. There are grayscale (8bit) as well as RGB and CMYK (24bit) colour based output formats available as well. No dithering applies on those patterns. Memory and processing time requirements are higher for those.

Image Processing Recommendations:

The choice which processing engine and pattern to use depends on the files, the output device and the client expectations. There is no “one answer fits all”. The user is encouraged to experiment with different options and do test prints. PDF Magick has a “Test” option in the Convert and Print screen that will test process 1 file out of the job on all 3 processing engines with their currently selected settings and report the processing speed result. The “Test” command will not Print to the device, it is a file processing speed test only.

**From a clients point of view; CONSISTENCY IS KEY.
It is better to do some test prints of representative files in advance;
ensure that they match client expectations and stay with that processing
method unless there is good reason to change.**



A test chart Pdf with fine lines, text, gray screens and images is available upon request.

Image Processing Workflow:

PDF Magick is very flexible and will not force a certain workflow style. Instead the design goal is to give options so that a team may choose a workflow style that suits the size of their operations the most.

There are 5 possible options when printing as image, 2 are **Task based** and 3 are **Queue based**. The schematics explain each option.

In general **Task based** workflows are recommended for small operations where 1 person is controlling 1 or more devices or where very close communication between 2 or more operators exist and there is clear understanding who is printing to the device.

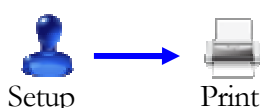
Queue based workflows are recommended for large operations where multiple setup operators queue print ready jobs to a central queue and a “Traffic Cop” or “Quarterback” person prints to multiple devices from a central queue.



TIP : The principles explained below apply to “Print as Image” but can equally be used for “Print as PDF” method; just by dropping the “Convert” step if the device is PDF capable. The icons show a single person, station or device but the principles can be applied to “many to many” scenarios as well.

The main goal of the suggested workflows below is to move the intense processing tasks to the most power ful PC and not tie up a print device during processing.

OPTION 1 – Task Based – “On the Fly”



Required Image files are created on the fly. This requires a PC that is powerful enough to process files faster then the device can print; otherwise you will experience stop & go behaviour. The image files that are created are not stored. If the job needs to printed again PDF Magick will process the files again. The benefit of this method is that there is no upfront processing time. As soon as the first page is processed the device will start printing and continue. Ensure that device configuration is set to “Print while processing” otherwise the device will wait until all files are processed. The print submission progressbar will gradually increase, depending on the processing speed.

OPTION 2 – Task Based – “Preprocess”



Required Image files are created in a temorary folder when operator processes with “Convert” option. Ensure that “Prepressing Mode” is enabled in config. This will create a link between the PDF and a matching Image. You will see (+tif) tags at the end of the filenames. The benefit of this method is that you will not tie up the print device if the PC cannot process files fast enough. Printing can be started after processing is completed. The print submission progressbar will increase very quickly.

OPTION 3 – Queue Based – “Preprocess”



Process files to Image with the “Convert” option and “Send” job to a print queue. Ensure that “Include printable Tif” option is selected. If you have multiple “versions” like fullsize and halfsize

you should convert for each version. PDF Magick will transfer the native PDF files as well as all required images into a /~tif/ subfolder in the queue. When it is time to print “Open” the job from the print queue and submit. The benefit of this method is that the print progress bar will advance very quickly and if reprints are required no additional processing is required. The print station will not be tied up for more then it needs to. If you have multiple powerful setup stations and average capacity print stations this workflow is the recommended option because the print station has very little work to do.

OPTION 4 – Queue Based – “On Demand”



Setup job and “Send” to the print queue. “Open” job from queue and “Convert” to a temporary local folder; it does not matter where. PDF Magick will create required images in the queue inside a subfolder of the job named /~tif/. A link between the PDF pages and the images will be established. The temporary files can be deleted after conversion is done. Now “Print” as image when required and the print progress bar will advance very quickly. The benefit of this method is to utilize a very powerful Print station but without tying up a device, especially if the print station is controlling multiple devices. Another benefit is not to do lengthy processing on setup stations if they are less powerful. As with all queue based workflows; if the job needs to be reprinted all images are ready. This option is also suitable if you are normally printing as PDF but the device has problems printing in PDF mode on a particular job. Note that the queue can be opened and Processing initiated from any available station; it does not have to be the print station if it is busy. PDF Magick will place the image files into the correct subfolder, no matter which station the job was opened from.

OPTION 5 – Queue Based – “On the Fly”



Setup job and “Send” to the print queue. “Open” job from queue and “Print as Image”. This will create the required images on the fly and store them in the queue job subfolder /~tif/ automatically. A powerful print station is required or you will experience stop & go. The benefit of this method is that you are not waiting for processing to finish; printing starts quickly even for very big jobs. Also there is no lengthy processing required on setup stations. Ensure that device configuration is set to “Print while processing” otherwise the device will wait until all files are processed. The print submission progressbar will gradually increase, depending on the processing speed. This option is also suitable if you are normally printing as PDF but the device has problems printing in PDF mode on a particular job.

Image Processing Sample 1:

Sample enlarged images below show the difference between a coarse and a fine pattern.

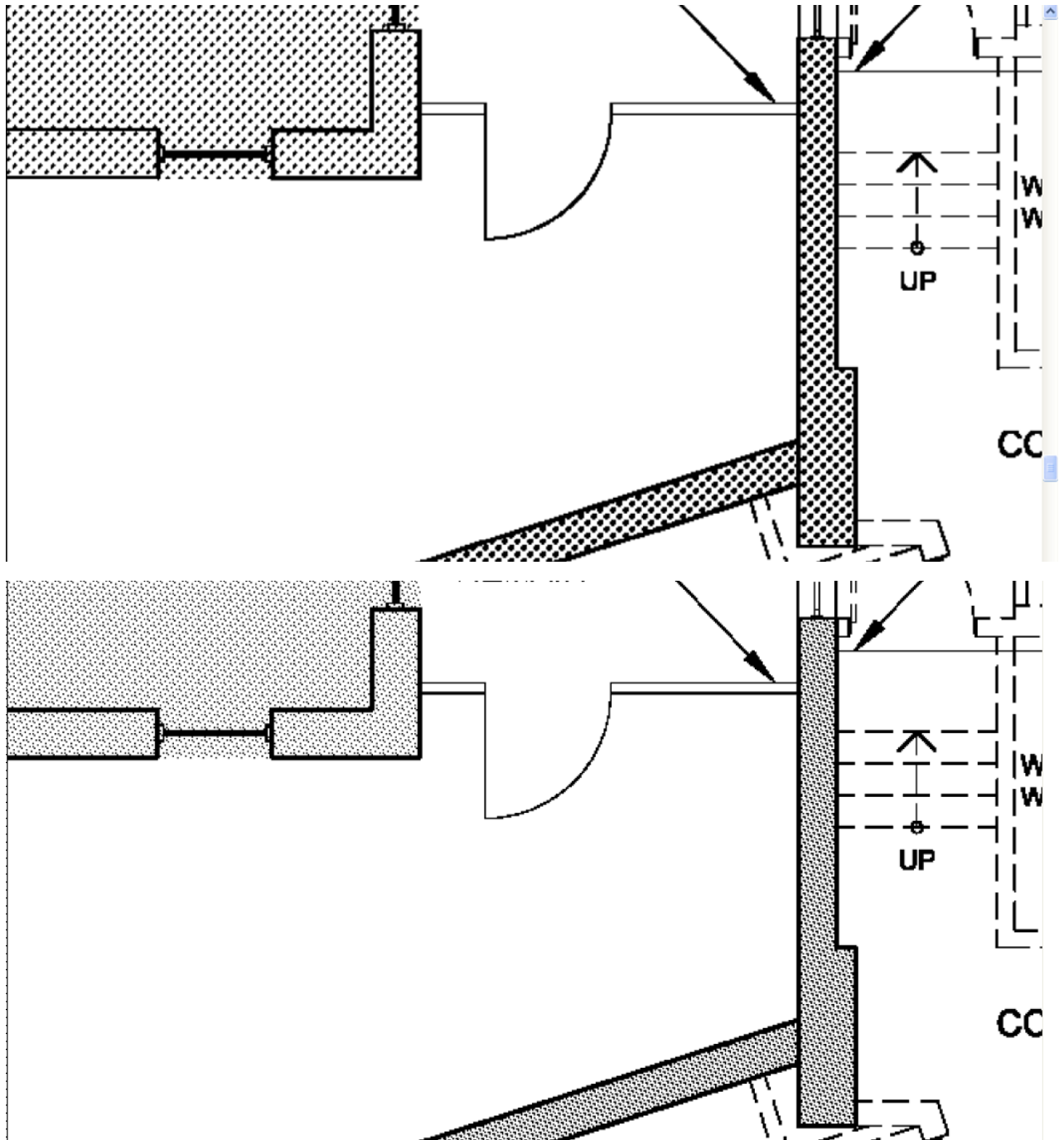


Image Processing Sample 2:

